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(54) **AEROSOL DISPENSING APPARATUS**

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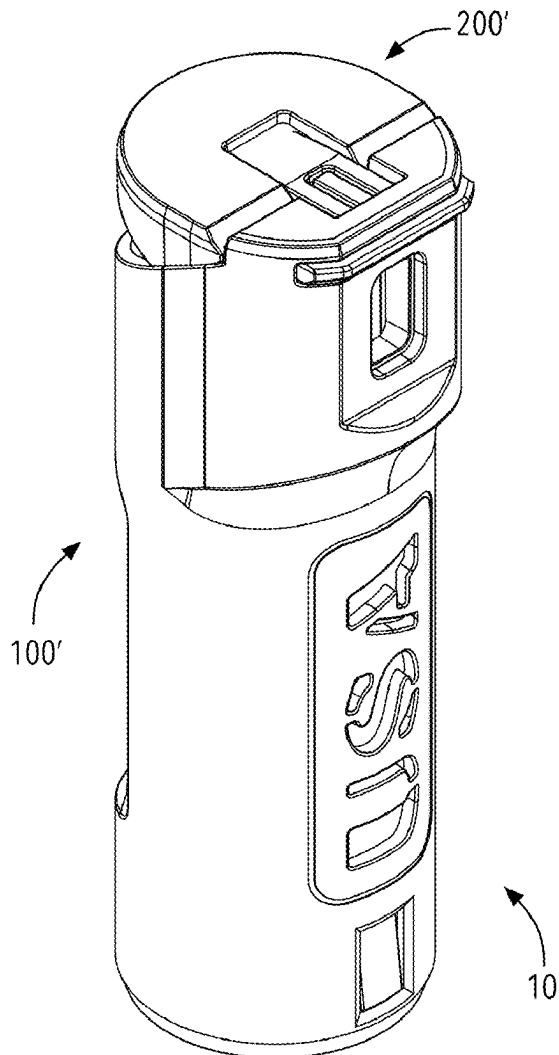
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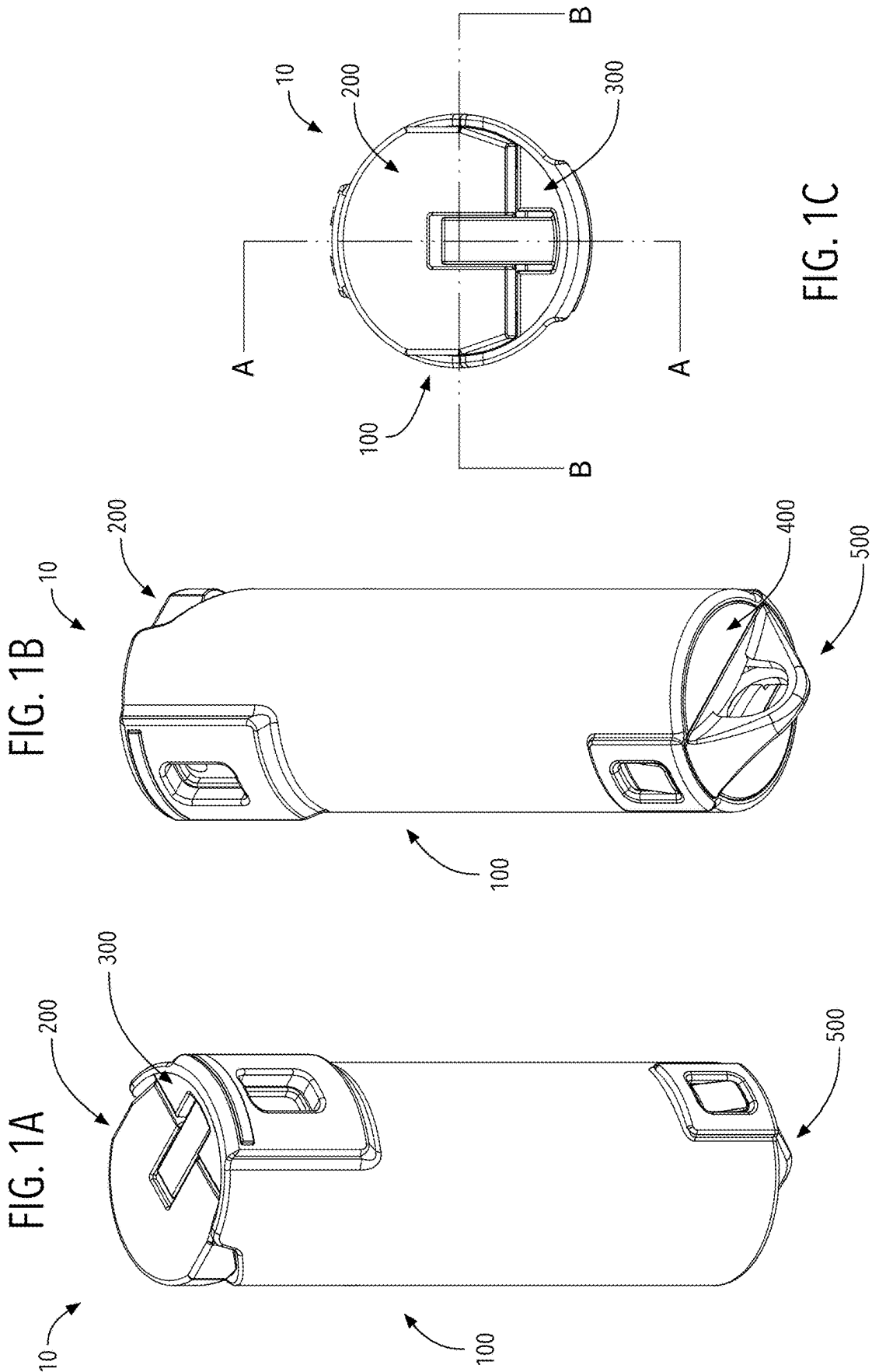
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(57)

ABSTRACT

A dispensing apparatus having a hollow body with a top opening and a bottom opening, the body including a discharge aperture therein, an actuator slidably arranged within the body and having a conduit therein, the conduit having a first opening and a second opening proximate a bottom thereof, a top cover secured to the body proximate the top, the cover having a movable section hingedly attached to the body, a leaf spring extending proximate the top opening and arranged to substantially abut a top surface of the movable section, thereby applying a downward force thereon, and a bottom cover removably secured to the bottom of the body, wherein the actuator is arranged to actuate an aerosol container within the body when downward force is 10 applied thereto, thereby applying force on a nozzle of the aerosol container and discharging therefrom through the conduit and out of the discharge aperture.





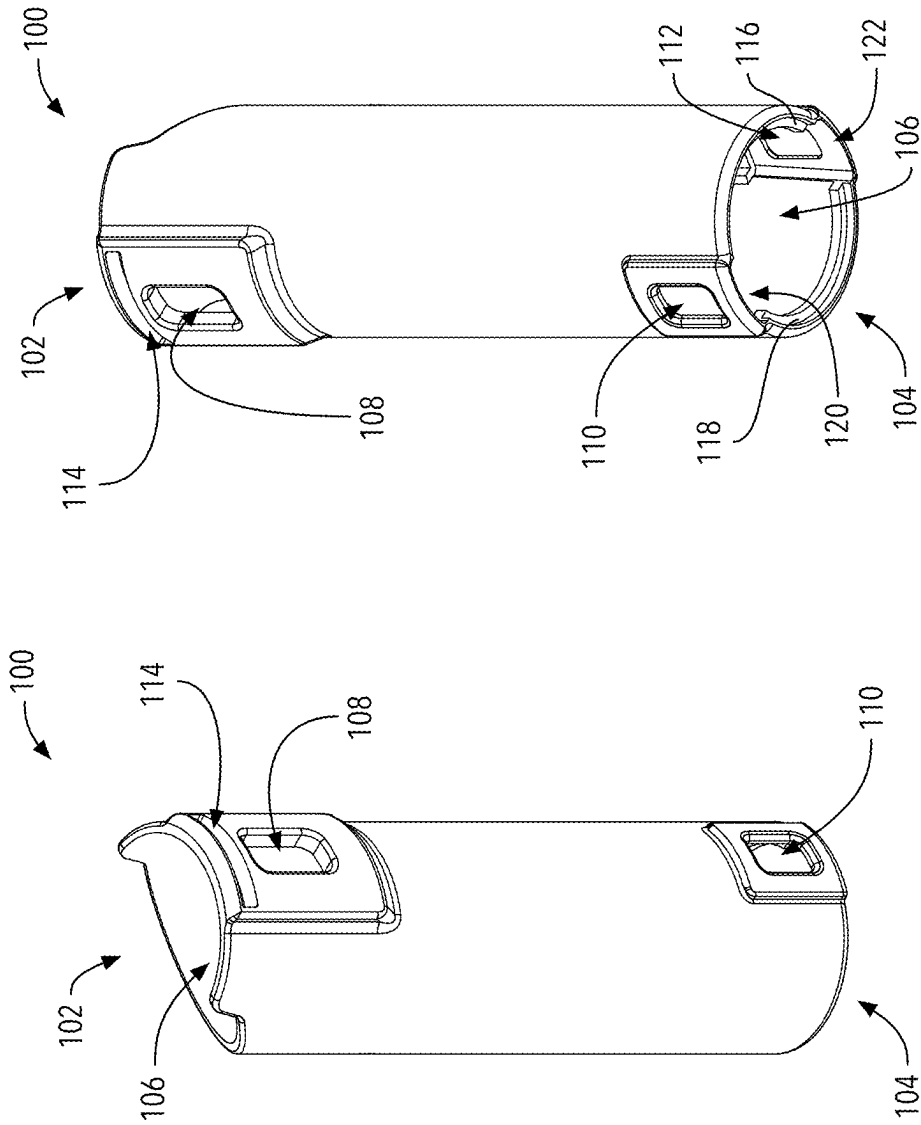


FIG. 2B

FIG. 2A

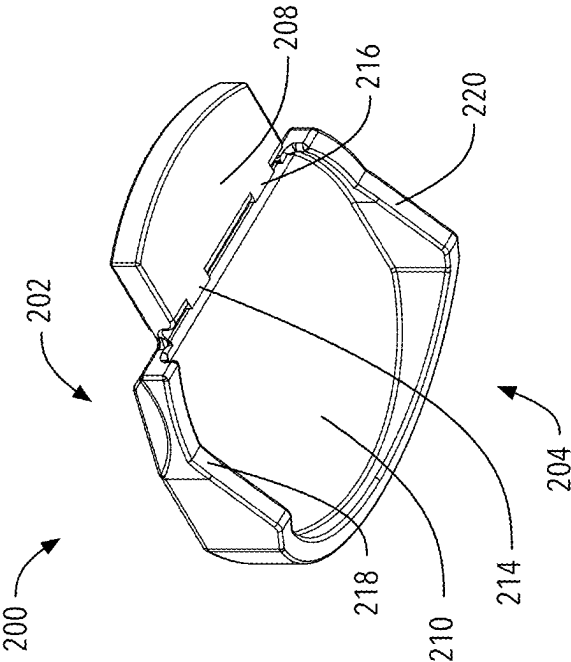


FIG. 3A

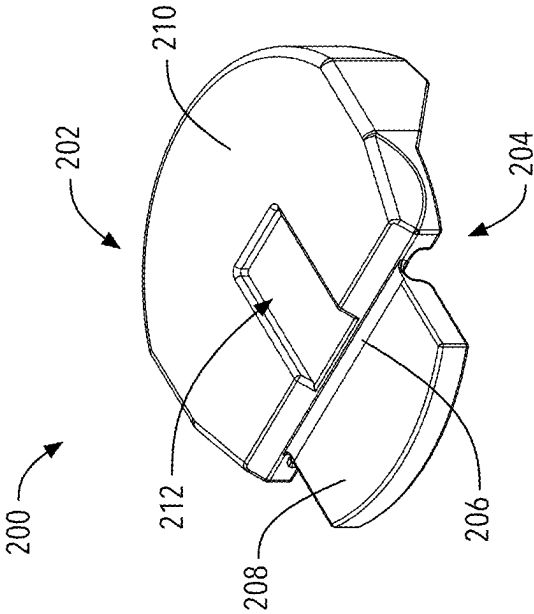


FIG. 3B

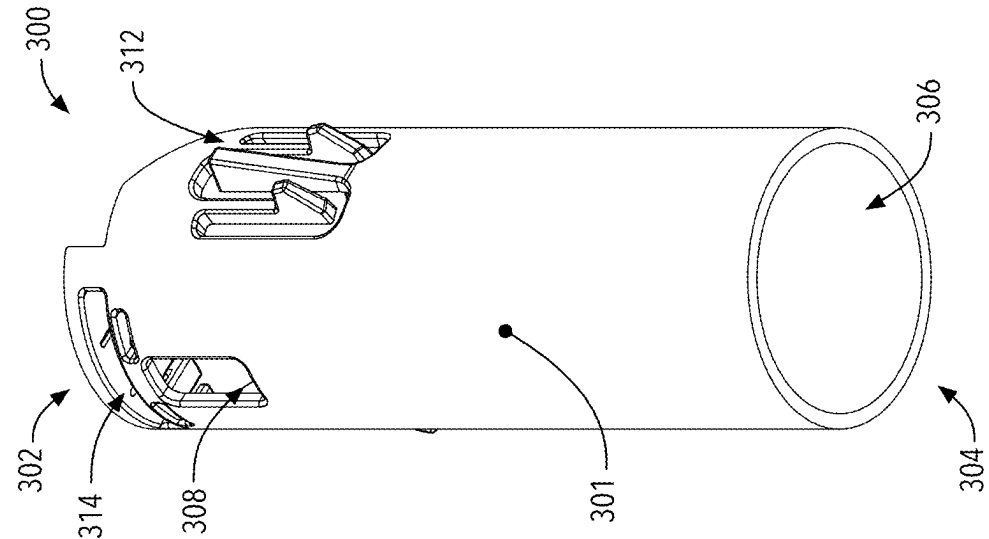


FIG. 4B

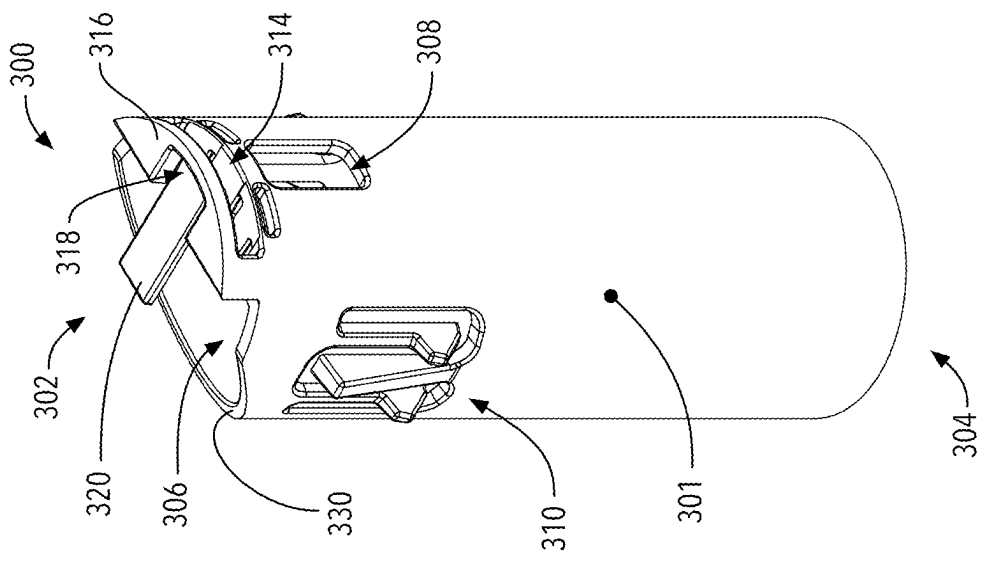


FIG. 4A

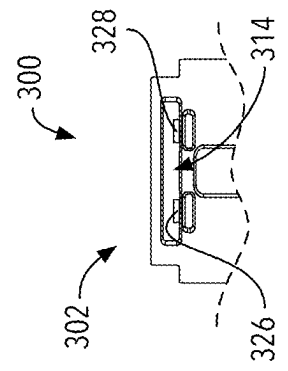


FIG. 4C

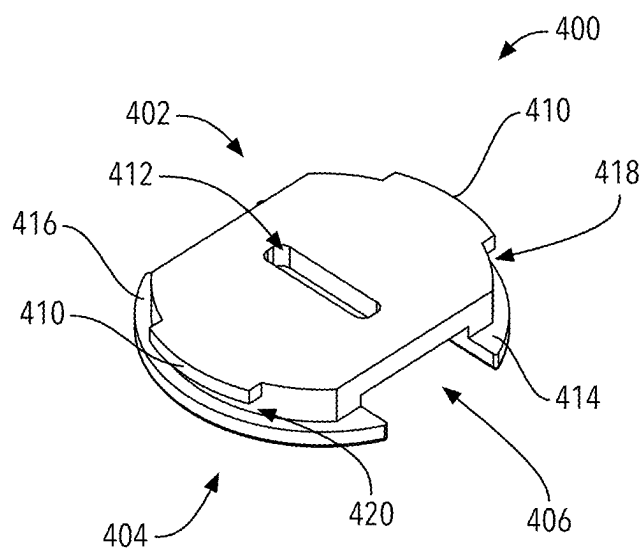


FIG. 5

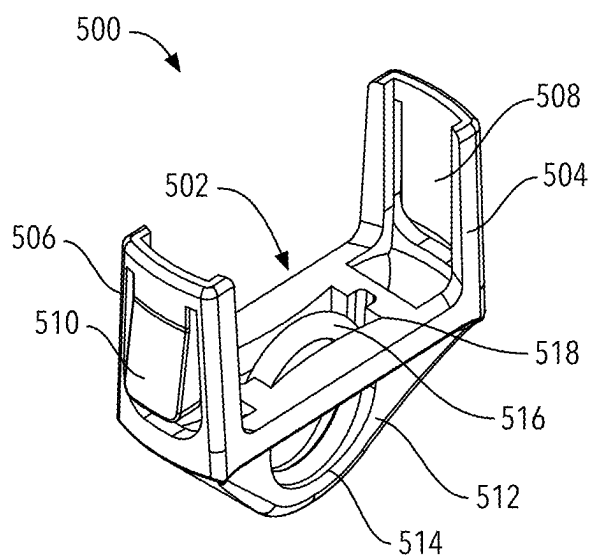


FIG. 6

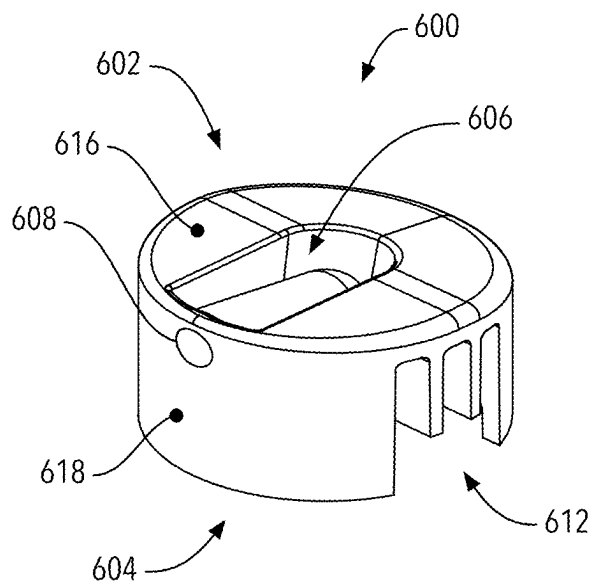


FIG. 7A

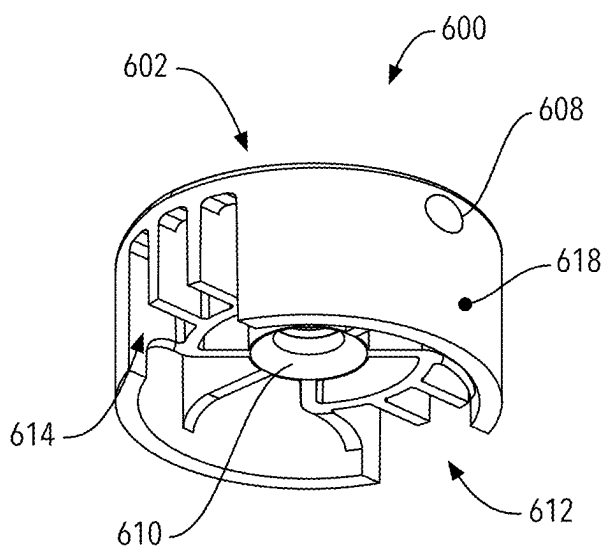


FIG. 7B

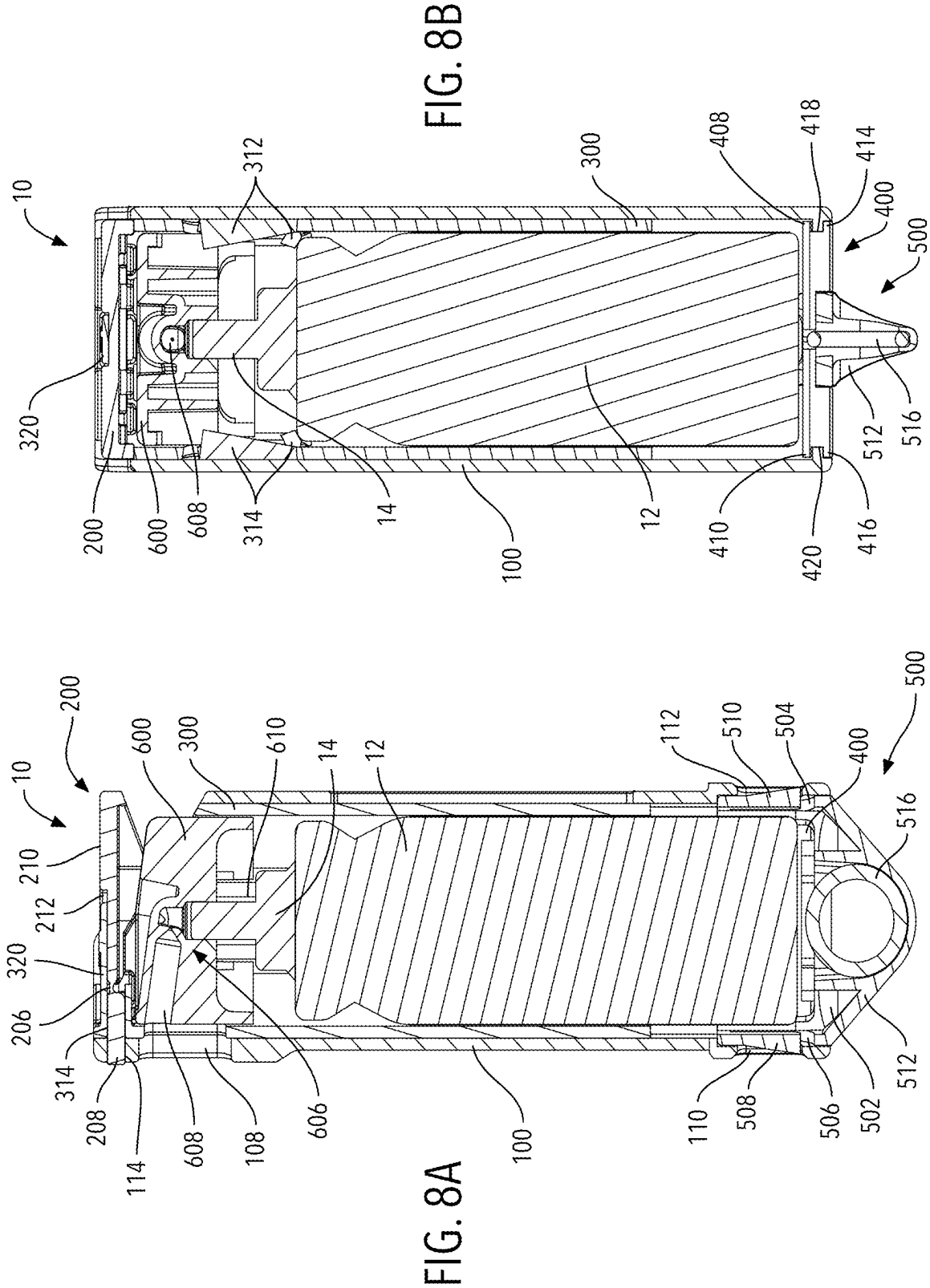


FIG. 9A

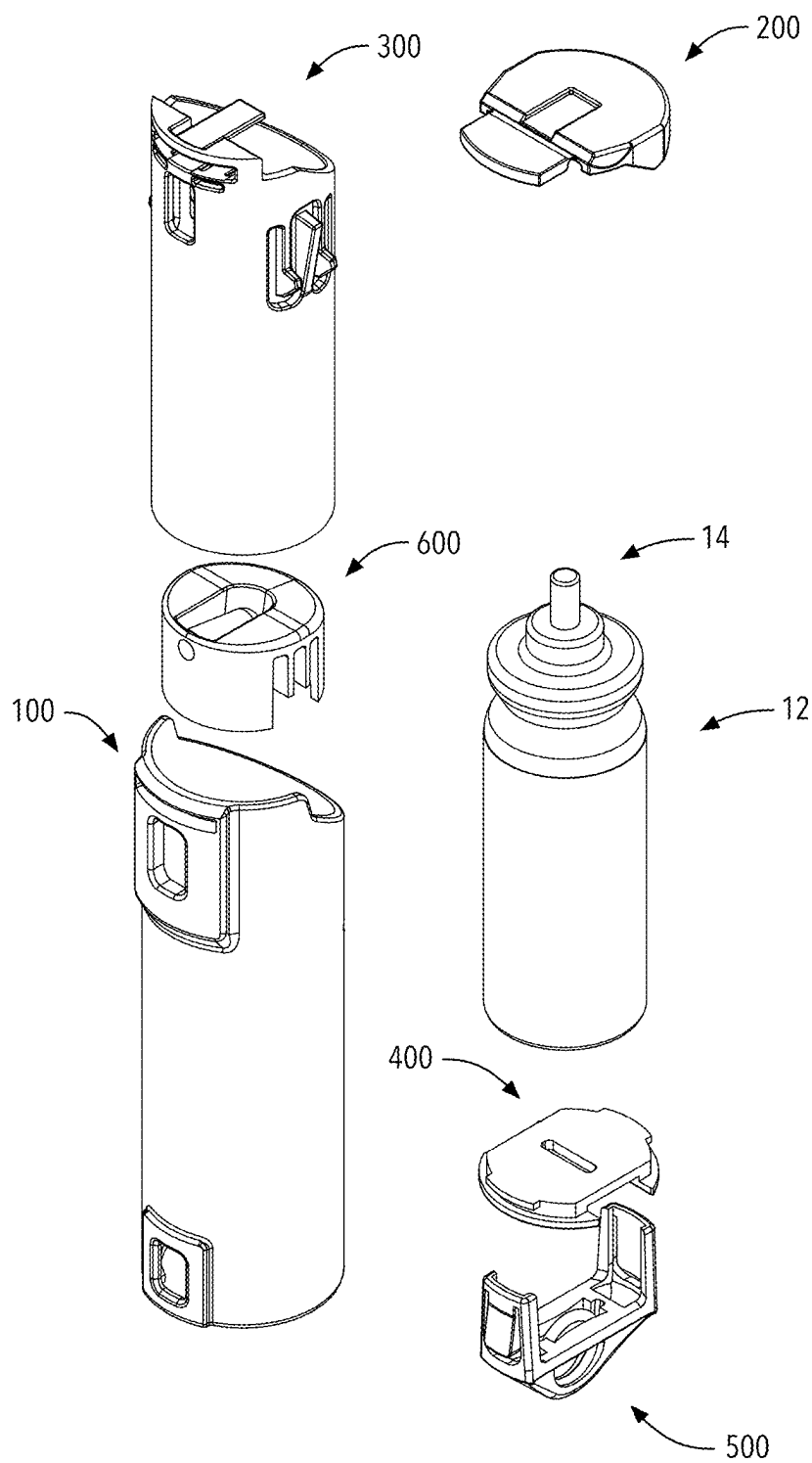
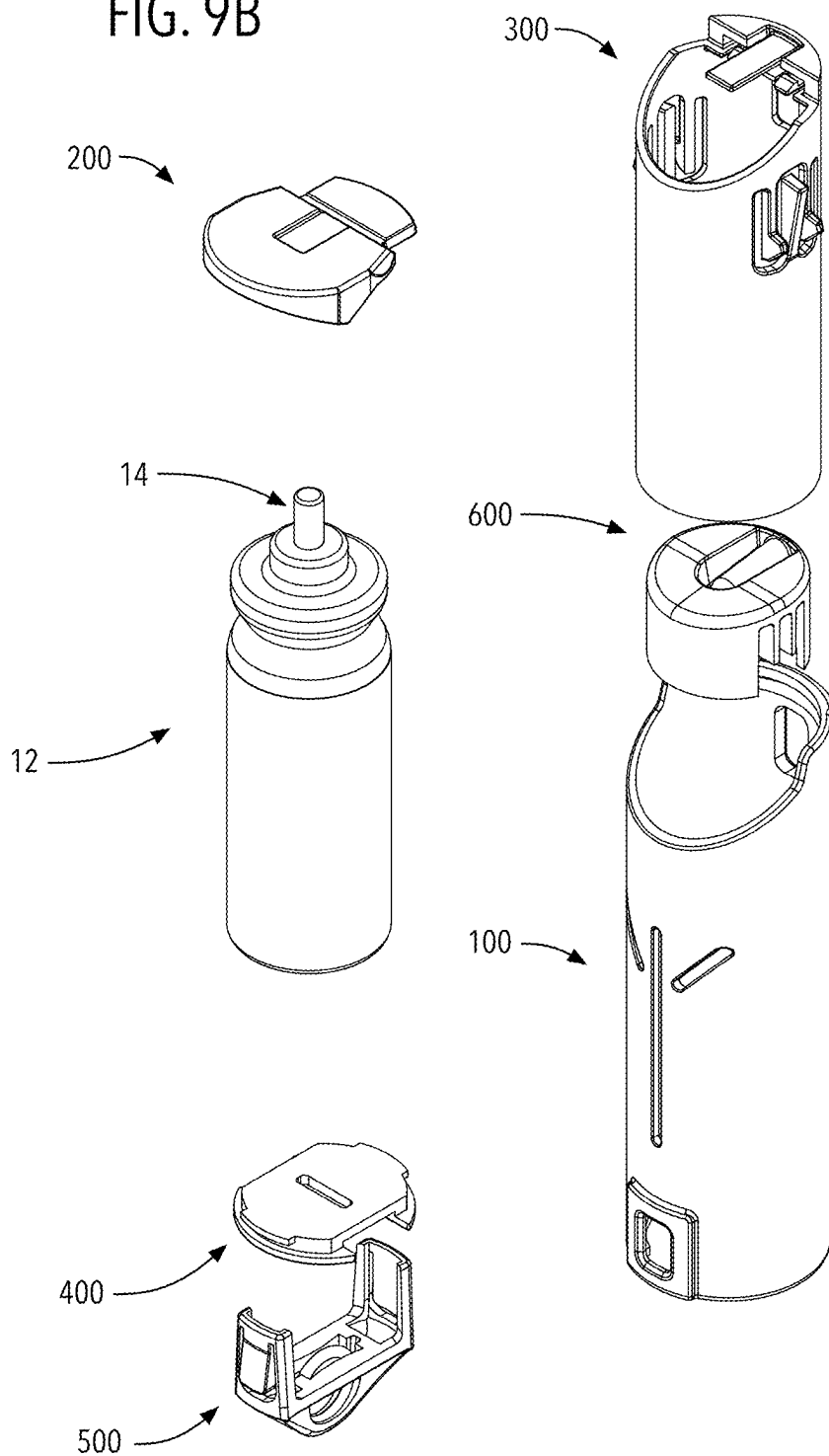


FIG. 9B



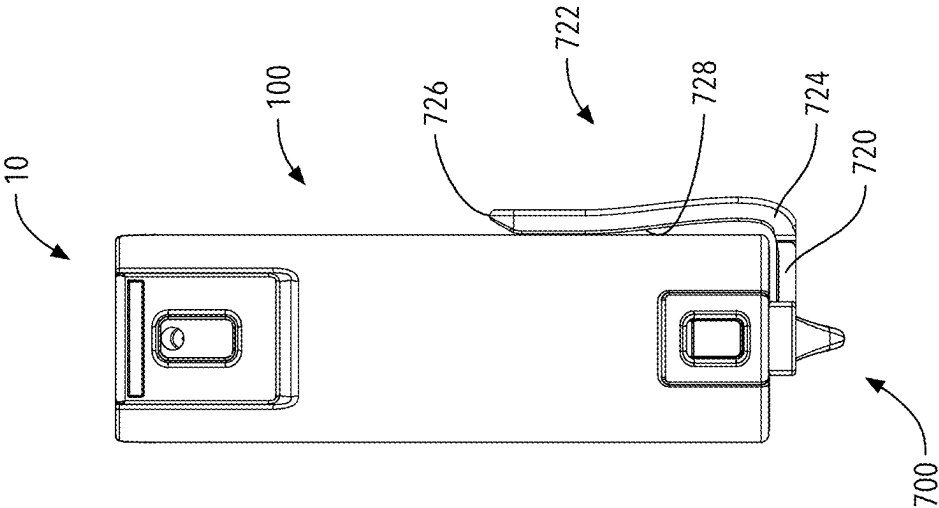


FIG. 10B

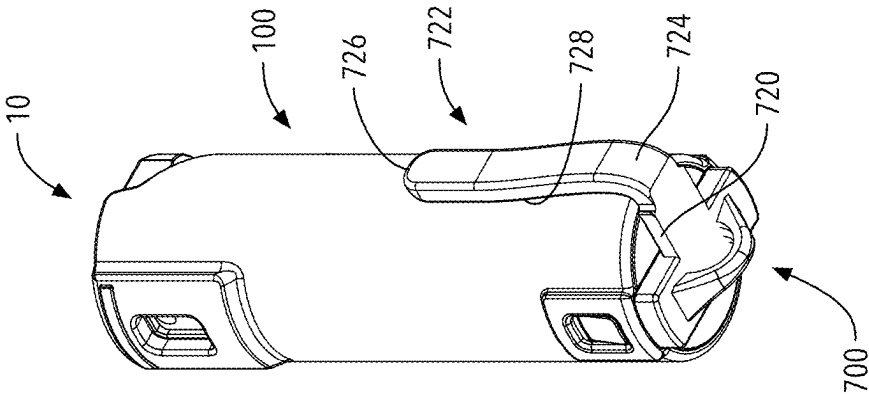


FIG. 10A

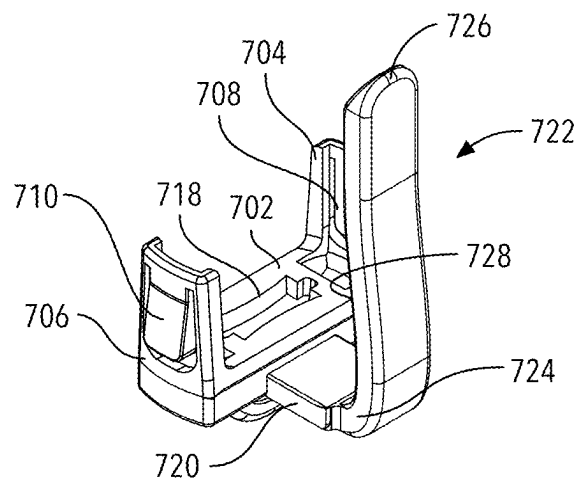


FIG. 11A

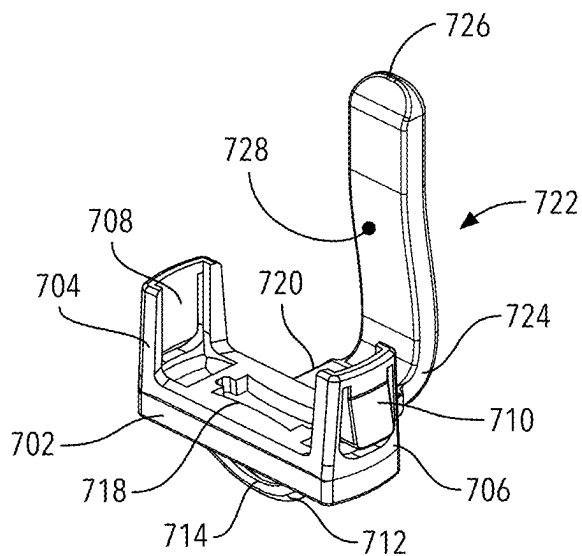


FIG. 11B

FIG. 12A

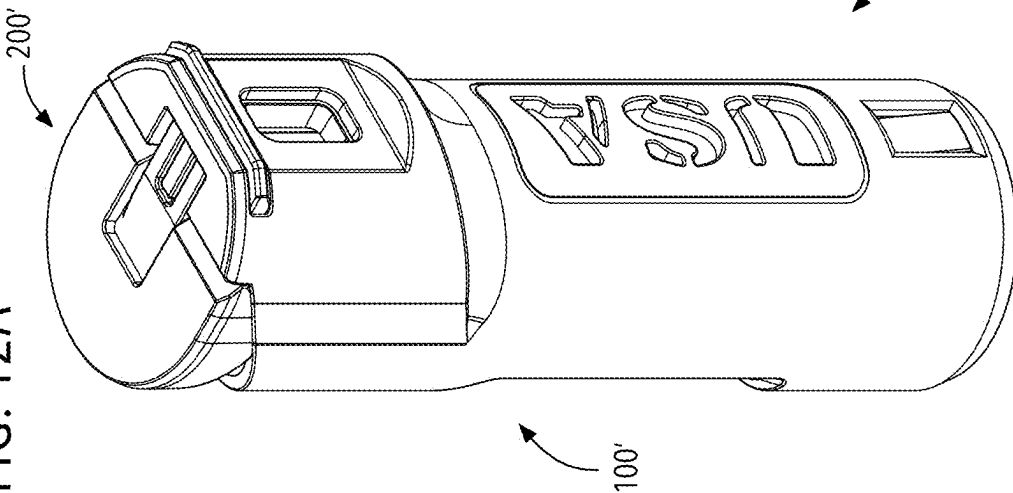


FIG. 12B

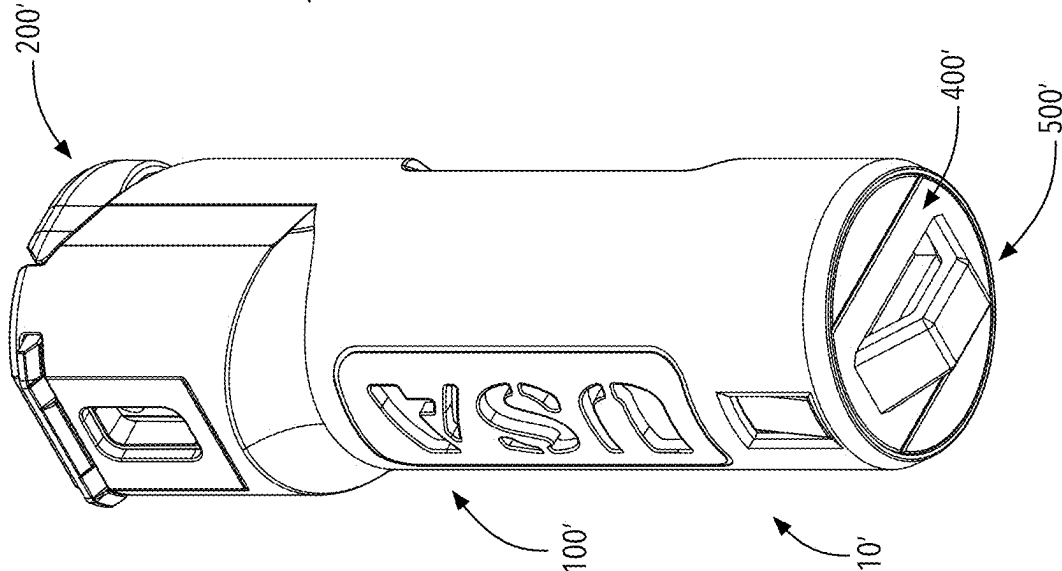


FIG. 13

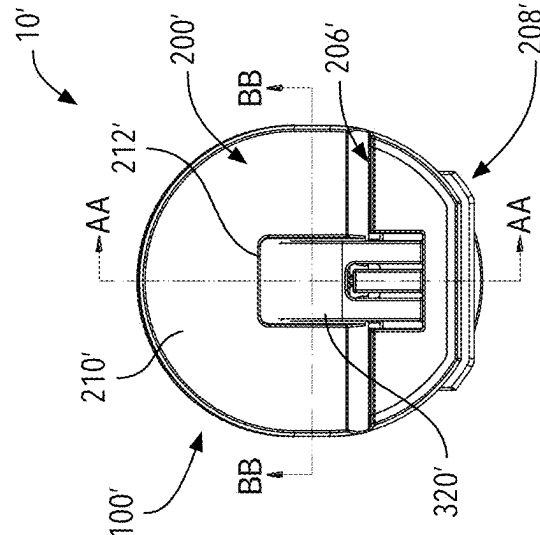


FIG. 14B

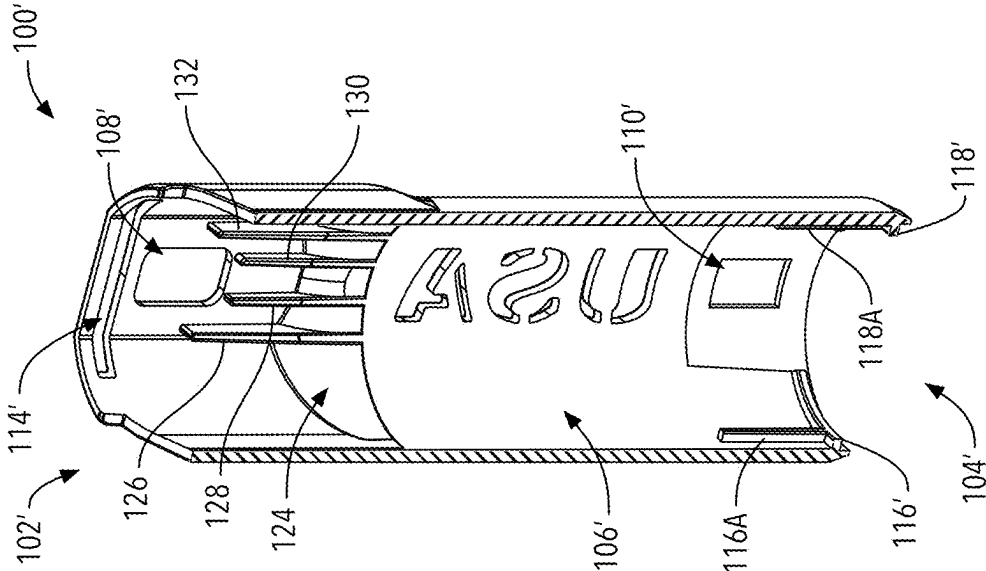


FIG. 14A

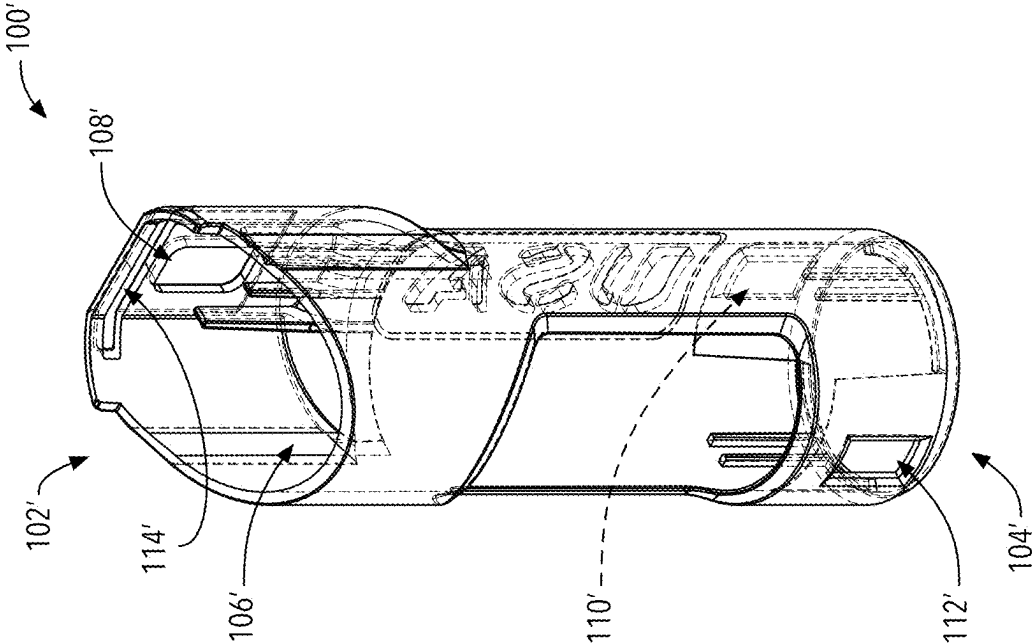


FIG. 15A

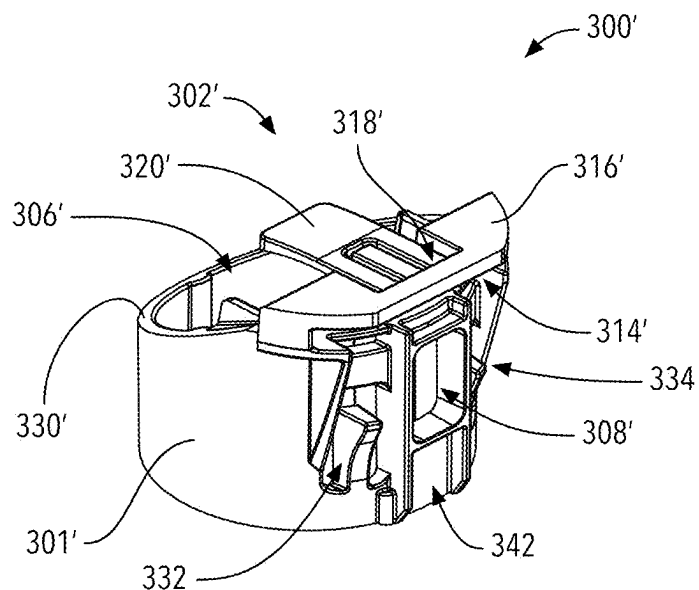


FIG. 15B

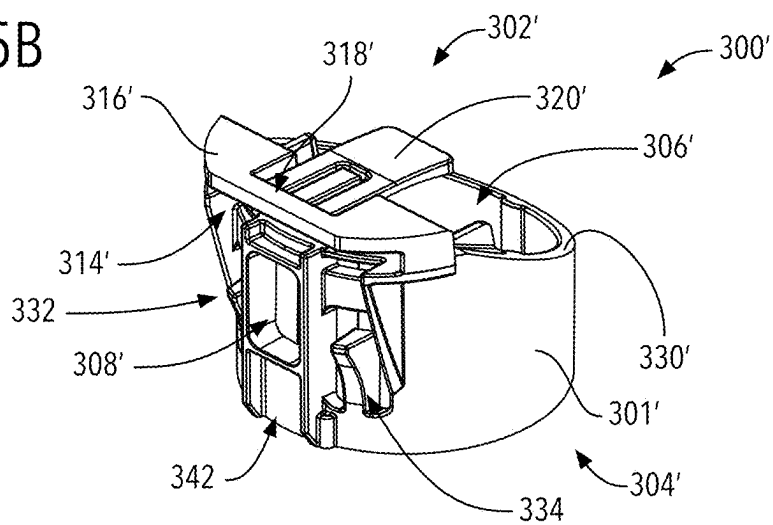
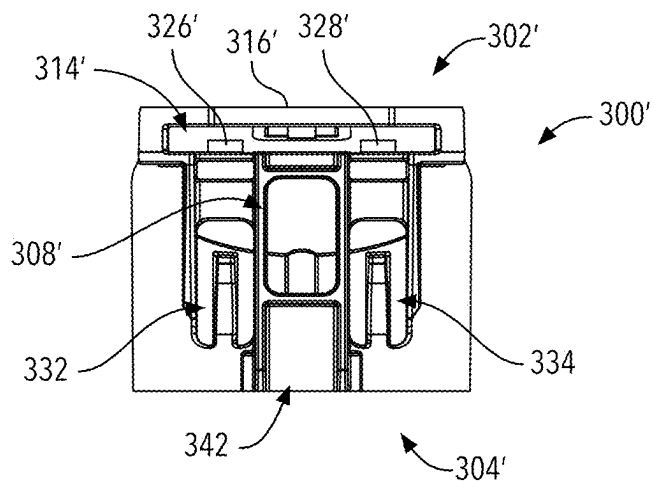


FIG. 15C



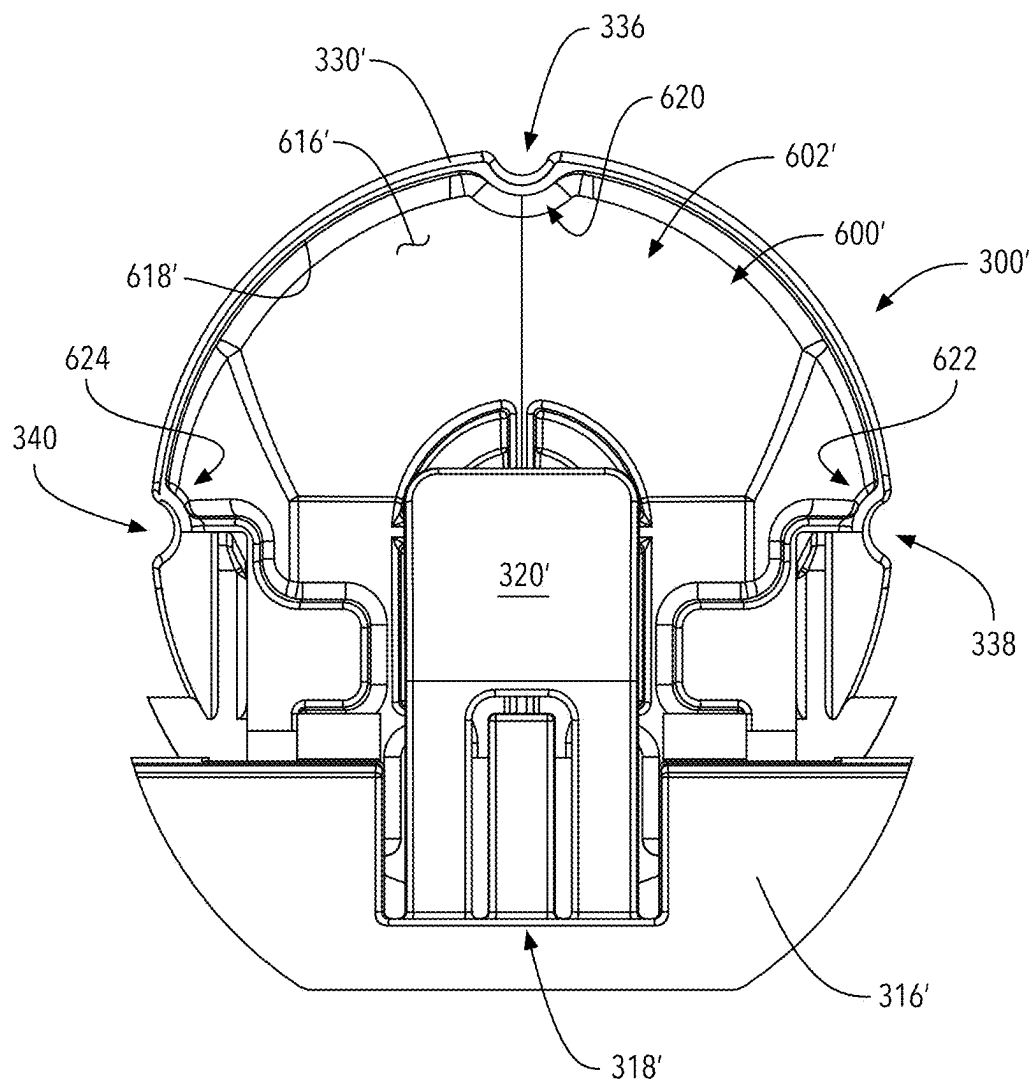


FIG. 15D

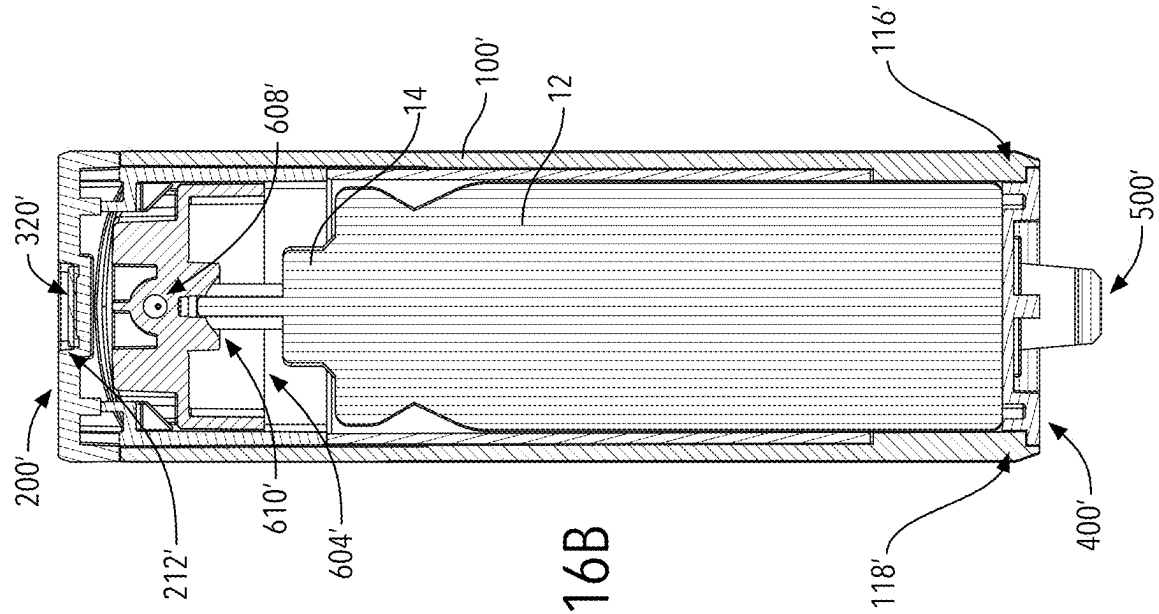


FIG. 16B

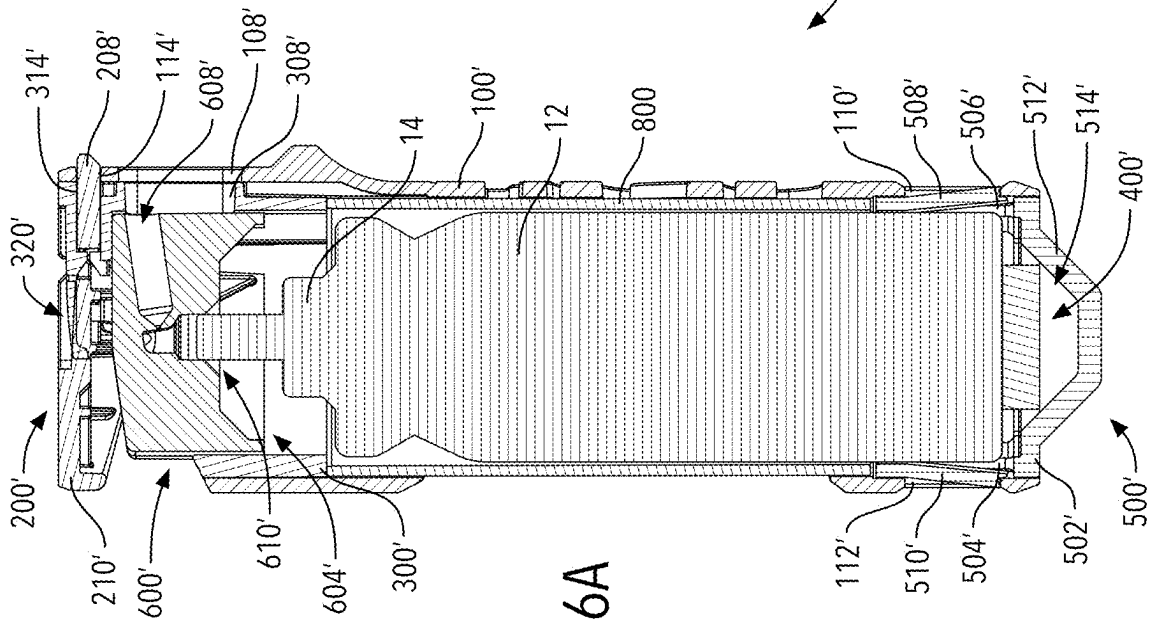
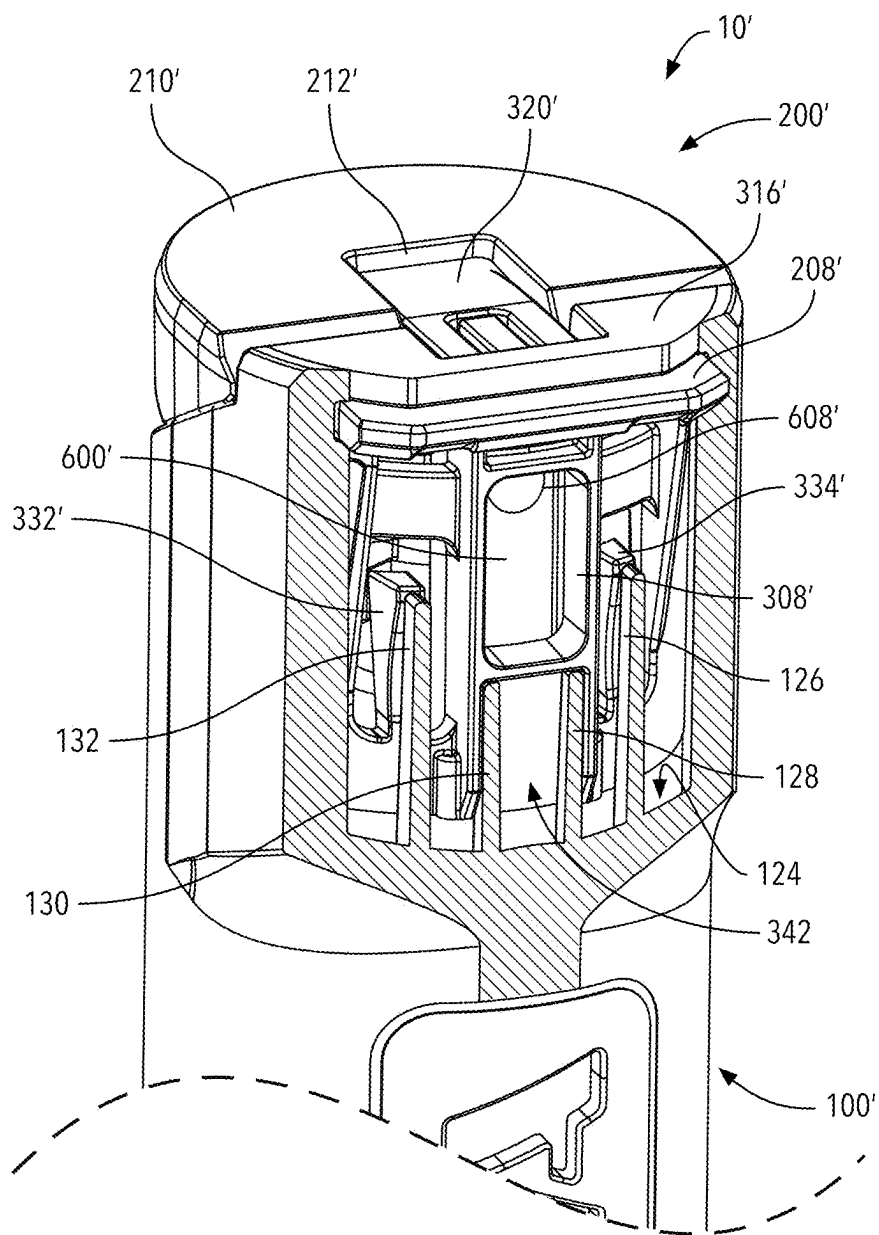


FIG. 16A

FIG. 17



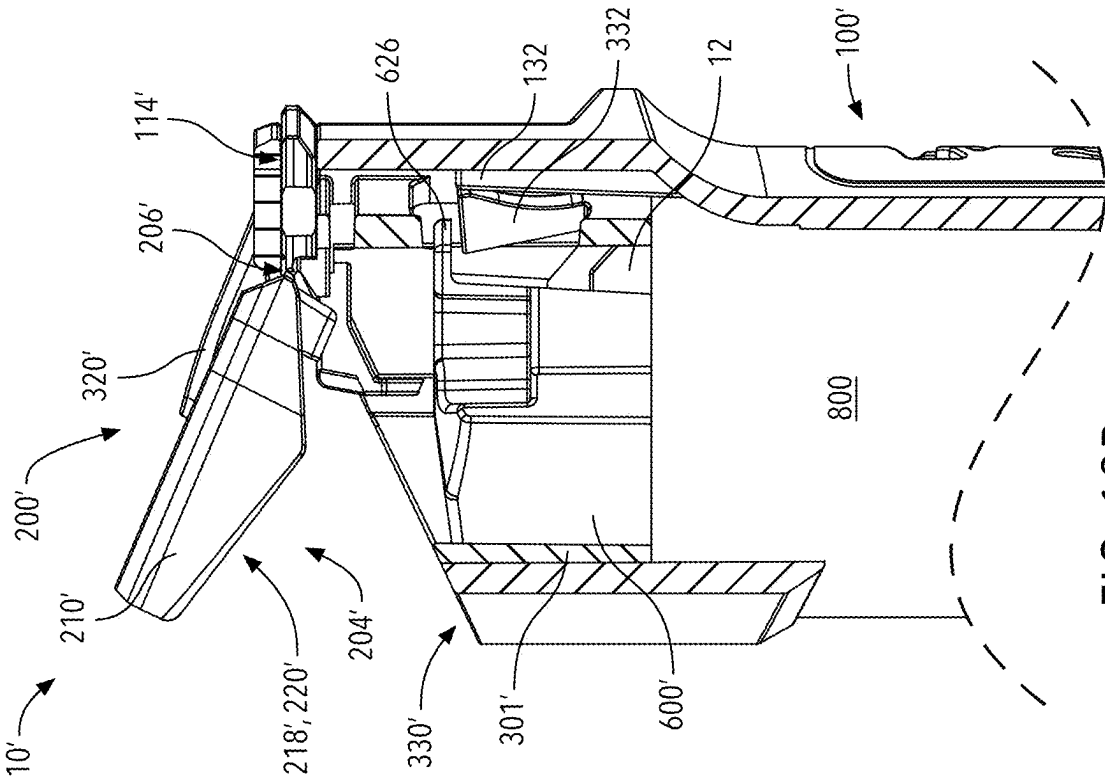


FIG. 18B

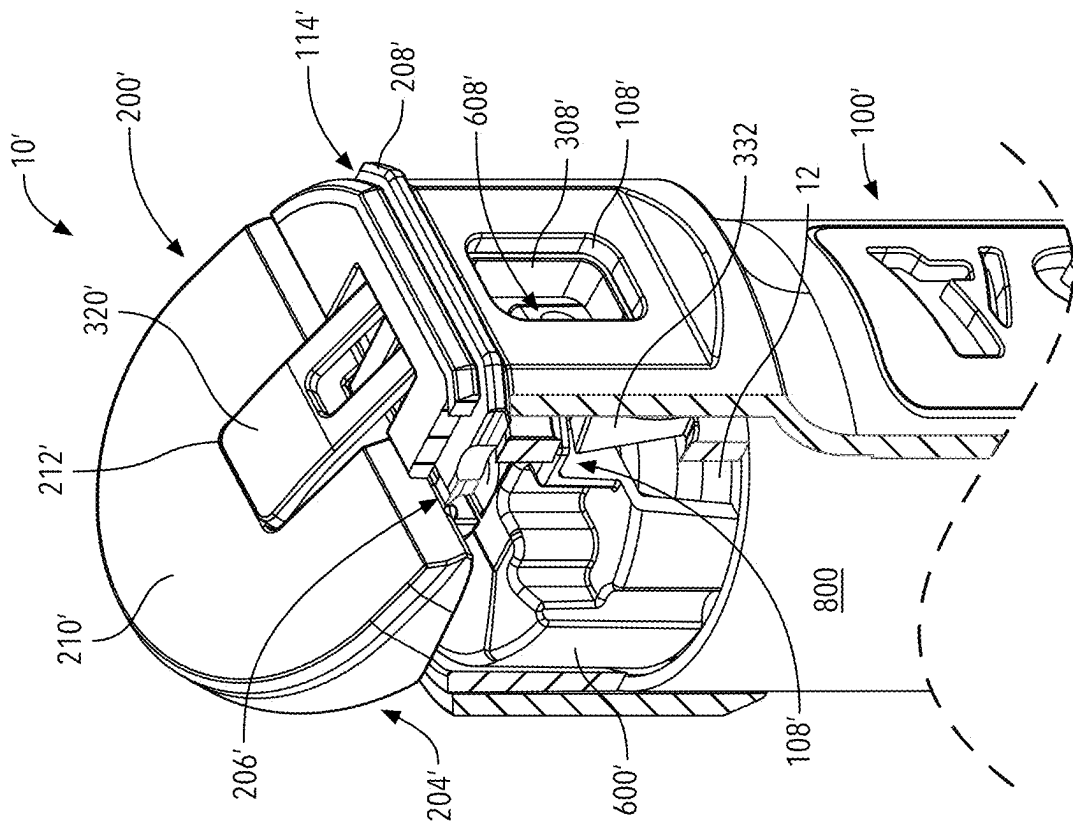


FIG. 18A

AEROSOL DISPENSING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority pursuant to 35 U.S.C. 119(e) to U.S. Provisional Application No. 63/556,565, filed Feb. 22, 2024, which application is incorporated herein by reference in its entirety.

FIELD

[0002] The present invention relates to an apparatus for an aerosol canister, and more specifically, to a system for a safety lid mechanism arranged at a top of the apparatus and a ring-release system of the apparatus.

BACKGROUND

[0003] Cases for aerosol or spray canisters typically include an actuator to engage the stem of the canister in order to dispel the canister's contents. There are a variety of aerosol dispensers that are used in many applications which include dispensing perfume, air fresheners, personal hygiene products, covering an article with a coat of paint, and dispensing cleaning products, amongst others. One specific application for an aerosol dispenser is as a personal defense device that, for example, directs a chemical repellant spray towards a potential human or animal threat.

[0004] Many cases for aerosol or spray canisters are adapted for personal use, such that are smaller in size, allowing them to be stored in purses or small compartments. In some uses, these canisters have "ring-systems" allowing them to be attached to a key ring-typically affixed to the case.

[0005] It is also typical for aerosol or spray canisters to have lids for dispensing actuators or cases for aerosol and spray canisters that are intended to either prevent accidental discharge or provide easy dispensing. With respect to personal defense devices, easy dispensing is crucial to ensure the safety of the user when a threat presents itself. However, the reason personal defense devices are effective is because their chemical contents are indiscriminately extremely painful to anyone who comes into contact with it.

[0006] Typically, lids or caps, are movably affixed to aerosol or spray canisters vis-à-vis an axial pin and torsion style springs. Although reliable in function, this structural arrangement also increases the overall size, rather, limits potential overall size reductions for personal or handheld aerosol or spray canisters.

[0007] Therefore, there is a long-felt need for a personal or handheld aerosol or spray canister having a reduced overall size, a release system for a key ring attachment or other attachment, a leaf spring system made of singular pieces which maintains a cover over an actuator to prevent accidental discharge, and an internal sleeve having movable members which restrict translational movement of one or more of an actuator within the sleeve or an aerosol canister within the sleeve.

SUMMARY

[0008] In some aspects, the present invention may comprise a dispensing apparatus having a body having an open top section and a bottom section, further comprising, at least one dispensing aperture and a first attachment aperture proximate the top section, a second and third attachment

aperture proximate the bottom section, a leaf spring hingedly attached to the top proximate the first attachment aperture, an actuator having a conduit defined by a dispensing opening and an inflow opening slidably arranged within the body, a top cover having a fixed section frictionally secured within the first attachment aperture and a movable section hingedly connected to the fixed section, a bottom cover rotatably secured to the bottom section of the body, and a quick-release attachment arranged to removably secure to the second and third attachment apertures, wherein an aerosol container is arranged to be removably secured within the body and dispenses contents from the dispensing aperture from the dispensing opening when the actuator is pushed towards the bottom section within the sleeve.

[0009] In some embodiments, the aforementioned dispensing apparatus may further comprise a sleeve arranged within the body, the sleeve having an open top section and a bottom section, the leaf spring hingedly attached proximate the top section, the sleeve having a second dispensing aperture, the sleeve having a fourth attachment aperture proximate the top section, wherein the second dispensing aperture is arranged to be in communication with the at least one dispensing aperture and the fourth attachment aperture is arranged to be substantially colinear with the first attachment aperture of the body when the sleeve is arranged within the body, wherein the actuator and the aerosol container are arranged within the sleeve.

[0010] In other possible embodiments, the sleeve further comprises a first plurality of fingers and a second plurality of fingers, each of the fingers hingedly connected to the sleeve, wherein both of the plurality of fingers restrict downward slidable movement of the actuator within the sleeve.

[0011] In further contemplated configurations, the sleeve further comprises a first finger and a second being disposed on opposite sides of the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.

[0012] In some other possible embodiments, the body includes a first external guide rail and a second external guide rail disposed on an internal surface thereof, the external guide rails adapted to bias each of the fingers of the sleeve in a radially inward direction when the sleeve is within the body.

[0013] In other possible configurations the body of the dispensing apparatus further includes a first internal guide rail and a second internal guide rail disposed on an internal surface thereof and arranged between the external guide rails, the internal guide rails arranged to rest at least partially within a channel of the sleeve, thereby restricting downward movement of the sleeve within the body.

[0014] In some embodiments, the leaf spring biases the movable section of the top cover downwardly thereby maintaining the movable section above the actuator.

[0015] In some contemplated configurations the quick-release attachment comprises a key ring extending therefrom, the quick-release attachment having a pair of protrusions arranged to frictionally and removably engage the second and third attachment apertures.

[0016] In further possible embodiments, the quick-release attachment comprises a clip extending therefrom and arranged to rest on an outer surface of the body, the quick-release attachment having a pair of protrusions

arranged to frictionally and removably engage the second and third attachment apertures.

[0017] In other aspects, the present invention may be a dispensing apparatus that includes a hollow body having a top opening and a bottom opening, the body including a discharge aperture proximate the top opening, an actuator slidably arranged within the body, the actuator having a conduit therein, the conduit having a first opening in communication with the discharge aperture and a second opening proximate the bottom, a top cover secured to the body proximate the top, the cover having a movable section hingedly attached to the body, a leaf spring extending from the top and arranged to substantially abut a top surface of the movable section, thereby applying a downward force thereon, and a bottom cover removably secured to the bottom, where the actuator is arranged to actuate an aerosol container within the body when downward force is applied to the actuator, thereby applying downward force to a nozzle of the aerosol container and discharging contents of the aerosol container through the conduit and out of the discharge aperture.

[0018] In other contemplated configurations, the leaf spring biases the movable section in a downward direction when the movable section is moved by a user in an upward direction, thereby substantially covering the actuator unless constant upward force is applied to the movable section.

[0019] In some possible embodiments, the aforementioned dispensing apparatus further comprises a quick-release attachment, the attachment having a pair of attachment tabs extending therefrom, the attachment tabs removably and frictionally engaged to a respective pair of attachment apertures arranged proximate the bottom opening of the body.

[0020] In further possible designs, the quick release-attachment includes at least one of: a key-ring; an aperture formed by a portion extending from the body; and, a spring clip.

[0021] In some embodiments, the aforementioned dispensing apparatus recited in Claim 11 further comprises: a hollow sleeve having a top opening and bottom opening, the sleeve having a second discharge aperture therein, the sleeve including one of: at least a first plurality of fingers hingedly secured thereon, wherein the sleeve is arranged to be inserted within the body and the actuator is arranged to be inserted within the sleeve, wherein at least one finger of the first plurality of fingers restricts translational movement of the actuator in a direction of the bottom opening of the sleeve; or, a first finger and a second being disposed on opposite sides of and proximate the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.

[0022] In further possible embodiments, the body includes a cover aperture proximate the top opening thereof and the sleeve includes a cover aperture proximate the top opening thereof, wherein the top cover includes a fixed portion arranged to be frictionally secured at least partially within the cover apertures, thereby securing the sleeve within the body.

[0023] These and other objects, features, and advantages of the present invention will become readily apparent upon a review of the following detailed description of the invention, in view of the drawings and appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] Various embodiments are disclosed, by way of example only, with reference to the accompanying schematic drawings in which corresponding reference symbols indicate corresponding parts, in which:

[0025] FIGS. 1A and 1B illustrate perspective views of the present invention;

[0026] FIG. 1C illustrates a top view of the invention shown in FIGS. 1A and 1B;

[0027] FIGS. 2A and 2B illustrate perspective views of body 100 of the present invention;

[0028] FIGS. 3A and 3B illustrate perspective views of cover 200 of the present invention;

[0029] FIGS. 4A and 4B illustrate perspective views of sleeve 300 of the present invention;

[0030] FIG. 4C illustrates a partial front view of sleeve 300 of the present invention;

[0031] FIG. 5 illustrates a perspective view of bottom cover 400 of the present invention;

[0032] FIG. 6 illustrates a perspective view a quick-release attachment 500 of the present invention;

[0033] FIGS. 7A and 7B illustrate perspective views actuator 600 of the present invention;

[0034] FIG. 8A illustrates a cross-sectional view taken generally along line A-A in FIG. 1C;

[0035] FIG. 8B illustrates a cross-sectional view taken generally along line B-B in FIG. 1C;

[0036] FIGS. 9A and 9B illustrate exploded views of the present invention;

[0037] FIGS. 10A and 10B illustrate perspective views of the present invention having quick-release attachment 700 installed therein;

[0038] FIGS. 11A and 11B illustrate perspective views of quick-release attachment 700 of the present invention;

[0039] FIGS. 12A and 12B illustrate perspective views of a second embodiment of the present invention;

[0040] FIG. 13 illustrates a top view of the invention shown in FIGS. 12A and 12B;

[0041] FIGS. 14A and 14B illustrate perspective views of body 100' of the invention shown in FIG. 12;

[0042] FIGS. 15A through 15C illustrate sleeve 300' of the invention shown in FIGS. 12A and 12B;

[0043] FIG. 15D illustrates cover 200' within sleeve 300' of the invention shown in FIGS. 12A and 12B;

[0044] FIG. 16A illustrates a cross-sectional view taken generally along line AA-AA in FIG. 13;

[0045] FIG. 16B illustrates a cross-sectional view taken generally along line BB-BB in FIG. 13;

[0046] FIG. 17 illustrates a partial front perspective view of the invention shown in FIGS. 12A and 12B with a frontal portion thereof cutaway; and,

[0047] FIGS. 18A and 18B illustrate a partial cutaway of the invention shown in FIGS. 12A and 12B.

DETAILED DESCRIPTION

[0048] At the outset, it should be appreciated that like drawing numbers on different drawing views identify identical, or functionally similar, structural elements. It is to be understood that the claims are not limited to the disclosed aspects.

[0049] Furthermore, it is understood that this disclosure is not limited to the particular methodology, materials and modifications described and as such may, of course, vary. It

is also understood that the terminology used herein is for the purpose of describing particular aspects only and is not intended to limit the scope of the claims. As such, those in the art will understand that any suitable material, now known or hereafter developed, may be used in forming the present invention described herein.

[0050] Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood to one of ordinary skill in the art to which this disclosure pertains. It should be understood that any methods, devices or materials similar or equivalent to those described herein or understood to one having skill in the art within, can be used in the practice or testing of the example embodiments.

[0051] The word “example” or “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations.

[0052] If the specification states a component or feature “may,” “can,” “could,” “should,” “would,” “preferably,” “possibly,” “typically,” “optionally,” “for example,” “often,” or “might” (or other such language) be included or have a characteristic, that a specific component or feature is not required to be included or to have the characteristic. Such a component or feature may be optionally included in some embodiments, or it may be excluded.

[0053] The terms “having,” “has,” “including,” “includes” are intended to be substantially synonymous with the terms “comprising” and/or “comprises” and/or “comprise”.

[0054] It should be appreciated that the term “substantially” is synonymous with terms such as “nearly,” “very nearly,” “about,” “approximately,” “around,” “bordering on,” “close to,” “essentially,” “in the neighborhood of,” “in the vicinity of,” etc., and such terms may be used interchangeably as appearing in the specification and claims. It should be appreciated that the term “proximate” is synonymous with terms such as “nearby,” “close,” “adjacent,” “neighboring,” “immediate,” “adjoining,” etc., and such terms may be used interchangeably as appearing in the specification and claims. The term “approximately” is intended to mean values within ten percent of the specified value.

[0055] It should be understood that use of “or” in the present application is with respect to a “non-exclusive” arrangement, unless stated otherwise. For example, when saying that “item x is A or B,” it is understood that this can mean one of the following: (1) item x is only one or the other of A and B; (2) item x is both A and B. Alternately stated, the word “or” is not used to define an “exclusive or” arrangement. For example, an “exclusive or” arrangement for the statement “item x is A or B” would require that x can be only one of A and B. Furthermore, as used herein, “and/or” is intended to mean a grammatical conjunction used to indicate that one or more of the elements or conditions recited may be included or occur. For example, a device comprising a first element, a second element and/or a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first

element, a second element and a third element; or, a device comprising a second element and a third element.

[0056] Moreover, as used herein, the phrases “comprises at least one of” and “comprising at least one of” in combination with a system or element is intended to mean that the system or element includes one or more of the elements listed after the phrase. For example, a device comprising at least one of: a first element; a second element; and, a third element, is intended to be construed as any one of the following structural arrangements: a device comprising a first element; a device comprising a second element; a device comprising a third element; a device comprising a first element and a second element; a device comprising a first element and a third element; a device comprising a first element, a second element and a third element; or, a device comprising a second element and a third element. A similar interpretation is intended when the phrase “used in at least one of: ” or “one of: ” is used herein.

First Exemplary Embodiment

[0057] Adverting now to the figures, FIGS. 1A and 1B generally illustrate perspective view of the present invention, aerosol dispensing apparatus **10** (hereinafter “apparatus”), fully assembled and FIG. 1C illustrates a top view of apparatus **10**.

[0058] As disclosed herein, apparatus generally comprises six (6) components, although it should be noted that various components may be combined in alternative configurations.

[0059] Generally, FIGS. 1A through 1C illustrate body **100**, top cover **200**, sleeve **300**, bottom cover **400**, quick-release attachment **500**. Within sleeve **300** and body **100** is actuator **600** (shown separated in FIGS. 7A and 7B). In operation, sleeve **300** also houses canister **12** (shown in FIGS. 9A and 9B), which is an canister arranged to have a pressured gas and/or vapor therein, such as but not limited to: pepper spray, mace, variations thereof (e.g., fogger, stream, gel, and foam embodiments), or other dispensable contents that are released from a canister upon the canister having pressure applied to an actuator.

[0060] The following description relates to possible individual components of apparatus **10** and are shown in FIGS. 2A through 7B individually.

[0061] FIGS. 2A and 2B generally illustrate body **100** of apparatus **10**. In some aspects, body **100** may include top **102** and bottom **104**, both of which are respective openings to internal space **106**. In other words, body **100** is hollow and includes two openings arranged at opposite ends thereof. In a preferred embodiment, discharge aperture **108** is arranged proximate top **102** and is open to internal space **106**. As shown in FIGS. 2A and 2B, discharge aperture **108** may be arranged within a section of body **100** that protrudes therefrom. First attachment aperture **110** and second attachment aperture **112** are preferably proximate bottom **104** and are open to internal space **106**. Like discharge aperture **108**, attachment apertures **110** and **112** may be arranged on respective sections of body **100** that protrude from first groove **120** and second groove **122**. Grooves **120** and **122** may be described as depressions arranged within an internal surface of body **100** and those depressions on the internal surface of body **100** protrude from the outer surface of the body **100**.

[0062] Similarly, third attachment aperture **114** may be arranged on the section of body **100** that protrudes therefrom and is positioned above discharge aperture **108** on the same

protruding section, and attachment aperture **114** is open to internal space **106**. Body **100** may include first protrusion **116** and second protrusion **118** extending from an internal surface of body **100**, i.e., within internal space **106**. Protrusion **116** and **118** may have an arcuate arrangement and are substantially divided by attachment apertures **110** and **112** and/or grooves **120** and **122**.

[0063] In some embodiments, body **100** may have a substantially circular cross-section, however, it should be noted that various alternative geometries are contemplated, e.g., triangular, square, or any other polygonal configuration, thus, elements of body **100** may have alternative configurations to accommodate different cross-sectional arrangements. In some configurations, the internal surface of body **100** may include shelves or abutments protruding into internal space **106**. These shelves or abutments may be respectively arranged above grooves **120** and **122**.

[0064] FIGS. 3A and 3B generally illustrate cover **200** of apparatus **10**. Cover **200** includes top **202** and bottom **204**, and hinge **206** which hingedly connects fixed section **208** and movable section **210**. It should be noted that “fixed” and “movable”, as used to designate the sections of cover **200** are in reference to their functional arrangement when apparatus is fully assembled and/or in use, and such designations should not be considered restrictive on the scope of the appended claims. Depression **212** is arranged within movable section **210** and generally proximate top **202**. First stop **214** and second stop **216** are protruding portions that extend from bottom **204** of fixed section **208**. Movable section **210** also includes first angled edge **218** and second angled edge **220** extending from a bottom thereof. Hinge **206** is preferably a living hinge, i.e., a flexure bearing, i.e., a thinner section of material made from the same material as the two “rigid” pieces it connects. In other words, sections **208** and **210** may pivot about hinge **206**. In a preferred arrangement, section **208** is fixed, therefore section **210** pivots about hinge **206** while section **208** remains stationary.

[0065] FIGS. 4A and 4B generally show sleeve **300** of apparatus **10**, and FIG. 4C illustrates a partial front view of sleeve **300**. In some arrangements, sleeve **300** includes top **302** and bottom **304**, both of which generally are openings as sleeve **300** is a hollow body. Top **302** and bottom **304** define openings to internal space **306**. In a preferred embodiment, discharge aperture **308** is arranged proximate top **302** and is open to internal space **306**. Sleeve **300** includes first plurality of fingers **310** and second plurality of fingers **312** arranged proximate top **302**. Each finger of both plurality of fingers **310** and **312** are hingedly secured to sleeve **300** and form respective slots open to internal space **306** therebetween. Each finger of both plurality of fingers **310** and **312** may be hingedly secured to sleeve either proximate top **302** and/or bottom **304**, preferably in an alternating manner as generally depicted in FIGS. 4A and 4B. In a preferred configuration, each finger of both plurality of fingers **310** and **312** apply radially outward force in response to radially inward force applied thereon to maintain the position generally shown in FIGS. 4A and 4B. Thus, when the fingers are pushed inwardly, i.e., towards internal space, resistance from each of fingers is applied in the inverse direction. In a preferred configuration, sleeve **300** includes top surface **316** proximate top **302**. Leaf spring **320** is connected to a portion of top surface **316** via hinge **318**. As shown in FIG. 4C, fourth attachment aperture **314** may have protrusions **326** and **328** arranged therein or at least proximate thereto.

Protrusions **326** and **328** are tabs which may flex in directions towards top **302** or bottom **304**. Preferably, the protrusions will each have an angled face in the direction facing away from fourth attachment aperture **314** and a hard edge in a direction facing fourth attachment aperture **314**.

[0066] FIG. 5 illustrates bottom cover **400** of apparatus **10**. In some embodiments, cover **400** generally includes top **402** and bottom **404** having channel **406** disposed therein. First protrusion **408** and second protrusion **410** radially extend from cover **400** and are arranged proximate top **402**. First rim **414** and second rim **416** radially extend from cover **400** and are arranged proximate bottom **404**. First protrusion **408** and first rim **414** form channel **418** therebetween and similarly, second protrusion **410** and second rim **416** form channel **420** therein between. As shown in FIG. 5, bottom cover **400** also includes key ring aperture **412** disposed therein.

[0067] FIG. 6 illustrates a quick-release attachment, e.g., key ring device **500**, of apparatus **10**. In some configurations, key ring device **500** includes body **502** having first extension **504** and second extension **506** extending upwardly therefrom, where first extension **504** has first protruding tab **508** hingedly secured thereto and second extension **506** has first protruding tab **510** hingedly secured thereto. Tabs **508** and **510** arranged such that respective portions thereof extend radially outward past extensions **504** and **506** and apply radially outward force in response to radially inward force applied thereon to maintain the position generally shown in FIG. 6. In some arrangements, body **502** may include a protruding portion extending therefrom, e.g., housing **512**. Housing **512** includes aperture **514** therein. In some embodiments, housing **512** is also arranged to accept keyring **516** therein via slot **518**, where slot **518** extends into housing **512** allowing keyring **516** may rest within housing **512**. Further, keyring **516** may have a portion thereof extending upwardly from slot **518** and away from body **502**.

[0068] FIGS. 7A and 7B illustrate perspective views of actuator **600** of apparatus **10**. Actuator **600** includes top **602**, bottom **606**, top surface **616**, sidewall **618**, and conduit **606** arranged therein. Conduit **606** is defined by two openings, dispensing opening **608** arranged within sidewall **618** and inflow opening **610** proximate bottom **606**. Dispensing opening **608** is in fluid communication with inflow opening **610**. In some configurations, sidewall **618** includes first plurality of slots **612** and second plurality of slots **614** arranged therein.

[0069] In some embodiments and generally shown in FIGS. 10A through 11B, apparatus **10** may comprise quick-release attachment **700**. Generally, quick-release attachment **700** may comprise all of the components of quick-release attachment **500** with the addition of a clip, allowing apparatus **10** to be removably attached to belt loop and the like. Quick-release attachment **700** includes body **702** having clip extension **720** extending therefrom. Clip body **722** is connected to clip extension **720** via curved portion **724**. Clip body **722** is bounded by curved portion **724** and end **726**. As shown in FIGS. 10A and 10B, clip body **722** rests on an outer surface of body **100** and functions like a known-in-the-art spring clip, i.e., clip body **722** applies constant force onto the surface of body **100**, allowing an object to be sandwiched therebetween. It should be noted that in alter-

native embodiments, quick-release attachment 700 may not include housing 712 and therefore aperture 714 with key-ring therein.

[0070] The following description should be taken in view of all of the aforementioned Figures, and specifically, FIGS. 8A through 9B. FIG. 8A illustrates a cross-sectional view of apparatus 10 taken generally along line A-A in FIG. 1C and FIG. 8B illustrates a cross-sectional view of apparatus 10 taken generally along line B-B in FIG. 1C. FIGS. 9A and 9B generally show exploded perspective views of apparatus 10.

[0071] To assemble apparatus 10, first actuator 600 is slid into internal space 306 of sleeve 300 such that actuator 600 is arranged therein proximate top 302. Then sleeve 300 is slid into internal space 106 of body 100. When sleeve 300 is inserted into internal space 106 a surface thereof applies inward radial force onto each finger of both first plurality of fingers 310 and second plurality of fingers 312, such that the fingers are forcibly bent into internal space 306 of sleeve 300. This action allows each of the respective fingers of first plurality of fingers 310 and second plurality of fingers 312 to at least partially nest within first plurality slots 612 and second plurality of slots 614 of actuator 600, thereby maintaining actuator within internal space 306 of sleeve 300. At least one finger of first plurality of fingers 310 and/or second plurality of fingers 312 will abut at least one slot of first plurality slots 612 and/or second plurality of slots 614, thereby limiting the distance actuator 600 may slide within internal space 306 of sleeve 300 in a direction towards bottom 304 thereof. Thus, the combination of first plurality of fingers 310 and/or second plurality of fingers 312, when bent inwardly into internal space 306, and first plurality slots 612 and second plurality of slots 614 functions as a translational distance restrictor of actuator 600.

[0072] Top cover 200 maintains sleeve 300 within body 100. First, sleeve 300 is slid within internal space 106 of body 100 until third attachment aperture 114 of body 100 and fourth attachment aperture 314 of sleeve 300 are substantially aligned, i.e., substantially colinear, thus discharge aperture 108 of body 100 and second discharge aperture 308 of sleeve 300 are also substantially aligned, i.e., substantially colinear. Fixed section 208 of top cover 200 is inserted into both third attachment aperture 114 of body 100 and fourth attachment aperture 314 of sleeve 300. In some embodiments, one or more of fourth attachment aperture 314 or third attachment aperture 114 includes at least one tab therein. As shown in FIG. 4C, in a preferred embodiment, fourth attachment aperture 314 includes protrusions 326 and 328 (i.e., tabs). Fixed section, or portion, 208 is inserted into attachment apertures 114 and 314 and protrusions 326 and 328 are flexed in a downward direction from stops 214 and 216 of fixed section 208 pushing into the angled faces of protrusions 326 and 328. Once fixed section is inserted far enough within at least attachment aperture 314, stops 214 and 216 no longer apply pressure on protrusions 326 and 328 allowing protrusions to return to an unflexed position where the hard edge of the protrusions abuts stops 214 and 216, preventing fixed section 208 from being removed from attachment aperture 314 and therefore also preventing fixed section 208 from being removed from attachment aperture 114.

[0073] Once top cover 200 is installed within sleeve 300 and body 100, leaf spring 320 rests within depression 212 of movable section, or portion, 210 of top cover 200. Leaf spring 320 rests, i.e., does not apply force, when positioned

at least partially within or on depression 212. When movable section 210 is lifted, i.e., pivoted about hinge 206 in a direction towards top 102 of body 100, leaf spring 320 applies force onto depression 212 thereby pivoting movable section 210 in a direction towards bottom 104 of body 100. This action keeps movable section 210, thereby cover 200, over actuator 600, so that accidental downward pressure on actuator 600 cannot be applied (i.e., preventing accidental discharge of the contents of canister 12). Additionally, angled edges 218 and 220 of movable section 210 are arranged to abut edge 330 of sleeve 300 to prevent unwanted downward moveable (i.e., (i.e., preventing accidental discharge of the contents of canister 12).

[0074] Similarly, movable section 210 needs force applied in the upward direction towards top 102 in order to expose top surface 616 of actuator 600. When movable section 210 is hinged and maintained in the upward position, force may be applied in a direction towards bottom 104 of body 100 on top surface of actuator 600, thereby applying downward force on nozzle 14 of canister 12. When nozzle 14 is depressed downwardly, i.e., towards bottom 104, pressurized contents are release therefrom, up into inflow opening 610 of conduit 606 and out of dispensing opening 608, discharge aperture 308 and discharge aperture 108. So long as downward pressure is applied to top surface 616 of actuator 600, pressurized contents from canister 12 will be dispensed until canister is emptied and/or depressurized.

[0075] At this point, body 100, sleeve 300, actuator 600, and cover 200 are assembled and canister 12 may be inserted within internal space 306 of sleeve 300 within body 100. Canister 12 is slid into internal space 306 until nozzle 14 is at least partially within inflow opening 610 of actuator 600. In some arrangements, at least one finger of first plurality of fingers 310 and second plurality of fingers 312, when bent radially inward within internal space 306 of sleeve 300, may contact a top portion of canister 12 to prevent canister 12 from sliding further in a direction towards top 302. Since nozzle 14 of canister 12 is arranged to dispense upon downward force applied thereon, the at least one finger prevents accidental discharge of nozzle 14 by preventing canister 14 from sliding further in a direction towards top 302. Unless bottom cover 600 is engaged to bottom 104 of body 100, canister 12 may freely slide out of internal space 306 of sleeve 300 proximate bottom 104.

[0076] Bottom cover 400 rotatably secured to bottom 102 of body 100. Within internal space 106 proximate bottom 102 is first protrusion 116 and second protrusion 118, both of which separated by first groove 120 and second groove 122. In some aspects, bottom 102, or the bottom end, of body 100 has a pair of lips partially circumscribed there-around and disposed substantially between grooves 120 and 122, thusly forming a pair of arcuate channels below protrusions 116 and 118 and between the lip of bottom 102. Thus, protrusions 408 and 410 of bottom cover 400 are first inserted into grooves 120 and 122. Then bottom cover 400 is rotated about 90° so that protrusions 408 and 410 are positioned within each of the arcuate channels and below protrusion 116 and 118. Bottom cover 400 thereby prevents canister 12 from sliding through bottom 304 of sleeve 300 and bottom 104 of body 100.

[0077] To finalize the assembly of apparatus 10, quick-release attachment 500 is removably attached to body 100 proximate bottom 104. Body 502 is first aligned within channel 406 of bottom cover 400 and key-ring 516 is aligned

with aperture 412. Then protruding tabs 508 and 510 of extensions 504 and 506 are inserted into grooves 120 and 122 of body 100 proximate bottom 104, thereby bending tabs 508 and 510 towards one another until tabs 508 and 510 are seated within first and second attachment apertures 110 and 112 of body 100, i.e., in a snap-fit and/or quick-release manner, allowing tabs 508 and 510 to release in a radial outward direction. To remove quick-release attachment 500, tabs 508 and 510 are pushed/bent towards one another so they are released from their respective attachment apertures, allowing quick-release attachment 500 to be removed from body 100. It should be noted that quick-release attachment 700 removably engages body 100 of apparatus 10 in substantially identical fashion as quick-release attachment 500.

Second Exemplary Embodiment

[0078] Adverting now to FIGS. 12 through 20B which generally illustrate a second embodiment of apparatus 10, e.g., apparatus 10'. At the outset, the notable difference between the embodiments is the configuration of the respective sleeves, e.g., sleeve 300 of apparatus 10 and sleeve 300' of apparatus 10', specifically, the selection of either a pair of a plurality of fingers or a pair of spaced fingers, e.g., plurality of fingers 310, 312 of apparatus 10 and fingers 332, 334 of apparatus 10'.

[0079] The following description should be taken in view of FIGS. 12A through 13. FIGS. 12A and 12B illustrate perspective views of a second embodiment of a dispensing apparatus, designated generally as 10', with top cover 200' in a closed position. Apparatus 10' comprises a body 100', top cover 200', a sleeve (not shown), and bottom cover 400'. Body 100' includes a dispensing aperture proximate top 102'. A first attachment aperture and second attachment aperture (not shown) are located proximate bottom 104' for receiving quick release attachment 500' which secures bottom cover 400' to body 100'. Top cover 200' is hingedly attached to body 100'. Arranged proximate bottom 104' of body 100' is first protrusion 116' and second protrusion 118', substantially similar to those of body 100 in FIG. 2B and FIG. 8B, which protrusions interact with bottom cover 400' (shown in FIG. 12B) to removably secure the bottom cover to body 100'. A notable difference between the protrusions of body 100 and those of body 100' is that first protrusion 116' and second protrusion 118' include vertical extensions 116A and 118A that protrude from the internal surface of the sidewall of body 100'. Vertical extensions 116A and 118A restrict vertical movement of a second sleeve within body 100', which sleeve is shown in FIGS. 16A and 16B.

[0080] FIG. 13 illustrates a top view of dispensing apparatus 10'. Top cover 200' includes fixed section 208' and movable section 210', hingedly connected at hinge 206'. Movable section 210' includes depression 212' which is designed to accept leaf spring 320' therein, where leaf spring 320' (of the sleeve within body 100') biases movable section 210' of top cover 200' in a downward direction. Fixed section 208' is designed to be inserted within body 100', thereby connecting top cover 200' to body 100'.

[0081] FIGS. 14A and 14B illustrate perspective views of body 100' of dispensing apparatus 10', where FIG. 14B is partially cutaway. Body 100' is generally cylindrical and includes an open top 102' and an open bottom 104', defining openings to an internal space 106'. Dispensing aperture 108' is positioned proximate to top 102', providing an opening for dispensing the contents of an aerosol container (not shown).

A first attachment aperture 114' is also located proximate to top 102' and is configured to receive a portion of a top cover (not shown), specifically, a fixed section of the top cover. Proximate bottom 104' are first attachment aperture 110' and second attachment aperture 112', configured to receive a quick-release attachment (not shown).

[0082] Within internal space 106', a pair of internal guide rails 128 and 130 extend from a curved portion 124 towards top 102'. These guide rails are positioned below first attachment aperture 114' and are configured to interface with a sleeve (not shown) that houses the aerosol container and actuator (neither shown). A pair of external guide rails, 126 and 132, also extend inwardly from curved portion 124 towards top 102'. These external guide rails are positioned above first attachment aperture 110' and second attachment aperture 112', respectively, and are configured to interface with fingers (not shown) of the sleeve. External guide rails 126 and 132 are longer than internal guide rails 128 and 130. Curved portion 124 transitions smoothly and defines a larger diameter portion of internal space 106' proximate top 102' as compared to the portion of body 100' proximate bottom 104' and beneath curved portion 124. First protrusion 116' and second protrusion 118' extend inwardly from bottom 104' in a circumferential manner. These protrusions aid in securing bottom cover 400' (not shown) to body 100'.

[0083] FIGS. 15A through 15C illustrate sleeve 300' of dispensing apparatus 10'. FIG. 15A illustrates a perspective view of sleeve 300', highlighting the three-dimensional relationship between leaf spring 320', fingers 332 and 334, and channel 342. FIG. 15B illustrates another perspective view of sleeve 300', rotated relative to the view shown in FIG. 15A. This view further emphasizes the three-dimensional shape of sleeve 300' and the arrangement of its features. FIG. 15C illustrates a front view of sleeve 300', providing a clearer depiction of protrusions 326' and 328' extending downward from an internal surface of attachment aperture 314', as well as the living hinges connecting fingers 332 and 334 to wall 330' adjacent to channel 342.

[0084] Sleeve 300' is generally cylindrical and includes an open top 302' and open bottom 304', which define openings to internal space 306'. A circumferential wall defines sleeve 300' extends between and connects top 302' and bottom 304' and has edge 330' arranged opposite from dispensing aperture 308'. Sleeve 300' further includes dispensing aperture 308', attachment aperture 314', leaf spring 320', first finger 332, second finger 334, and channel 342. Dispensing aperture 308' is proximate top 302' and provides a passage for dispensing the contents of an aerosol container (not shown).

[0085] Attachment aperture 314' is proximate top 302' and includes protrusions 326' and 328' that extend downward, as best shown in FIG. 15C. These protrusions are configured to interact with a top cover (not shown), substantially identical those present in apparatus 10, described supra. Leaf spring 320' is proximate top 302' and is hingedly attached to sleeve 300' via hinge 318', which is preferably a living hinge. The free end of leaf spring 320' extends over top 302'. First finger 332 and second finger 334 are located dispensing aperture 308'. Each finger, 332 and 334, is hingedly attached to wall 330' by respective living hinges. The living hinges allow fingers 332 and 334 to flex inwardly towards internal space 306'. Channel 342 is disposed on the exterior surface of wall 330', opposite dispensing aperture 308'. This channel is configured to interact with guide rails (not shown) on the body (not shown) of the dispensing apparatus.

[0086] With reference to FIG. 15A through 15D, FIG. 15D illustrates a top plan view of actuator 600' positioned within sleeve 300'. Actuator 600' is generally cylindrical and includes top surface 616', sidewall 618', medial guide depression 620, first side guide depression 622, and second side guide depression 624. Medial guide depression 620 is centrally located and arranged within a respective external surface, while first side guide depression 622 and second side guide depression 624 are positioned on opposite sides of medial guide depression 620. These depressions are configured to interact with corresponding protrusions on sleeve 300' to maintain alignment during translation in the vertical direction and restrict horizontal/rotational movement of the actuator within the sleeve.

[0087] Sleeve 300' in FIG. 15D shows edge 330', leaf spring 320', first side guide protrusion 338, second side guide protrusion 340, and medial guide protrusion 336. Top surface 316' is shown and has living hinge 318' of leaf spring 320'. Leaf spring 320' is positioned substantially between top surface 316' (proximate the dispensing aperture) and edge 330' and is configured to bias a top cover (not shown) in a closed position. First side guide protrusion 338 and second side guide protrusion 340 extend radially inward and are positioned to interface with the corresponding side guide depressions, 624 and 622, on actuator 600'. Medial guide protrusion 336 extends inwardly from edge 330' and is positioned to interface with medial guide depression 620 of actuator 600'. This arrangement of protrusions and depressions ensures that actuator 600' remains centered within sleeve 300' during vertical translational movement and also restricts horizontal/rotational movement within sleeve 300'.

[0088] Comparing sleeve 300 (shown in FIGS. 4A through 4C) with sleeve 300' (shown in FIGS. 15A through 15C) reveals a key difference in the number and arrangement of the fingers that interact with the actuator. This design difference impacts how each sleeve interacts with the corresponding actuator (600 and 600', respectively).

[0089] Sleeve 300 includes first plurality of fingers 310 and second plurality of fingers 312. These fingers are arranged circumferentially around the sleeve proximate the top opening and are configured to flex inwardly when the sleeve is inserted into body 100 of dispensing apparatus 100. The plurality of fingers, 310 and 312, on sleeve 300 are designed to interact with the multiple slots on actuator 600.

[0090] In contrast, sleeve 300' includes only two fingers, 332 and 334, located on opposite sides of the sleeve proximate dispensing aperture 308' and within sidewall 301'. These fingers are each hingedly attached to sidewall 301' by a living hinge. This simplified design is intended to simplify the manufacturing process and reduce the potential of breakage. Additionally, sleeve 300' is designed to interact with medial guide depression 620, side guide depression 622 and second side guide depression 624 of actuator 600' via its corresponding protrusions (336, 338, and 340). This configuration with fewer interacting elements reduces complexity while providing precise alignment and controlled movement of the actuator within the sleeve.

[0091] Comparing actuator 600 (shown in FIGS. 3A and 3B) with actuator 600' (shown in FIG. 15D) reveals key differences in their structure and interaction with other components of the dispensing apparatus. Actuator 600 is designed to interact with a sleeve (300) having a plurality of fingers, while actuator 600' interacts with sleeve 300' having

only two fingers (332, 334). This difference is also reflected in the design of the guide depressions on actuator 600'.

[0092] Actuator 600 includes first plurality of slots 612 and second plurality of slots 614 arranged circumferentially around its sidewall (618). These slots are configured to receive and interact with the plurality of fingers of sleeve 300, restricting downward movement of the actuator.

[0093] In contrast, actuator 600' includes medial guide depression 620, first side guide depression 622, and second side guide depression 624. These depressions interface with medial guide protrusion 336, first side guide protrusion 338, and second side guide protrusion 340 of sleeve 300', respectively. This configuration with fewer, more precisely positioned interacting elements allows for controlled movement of actuator 600' within sleeve 300' while still providing stability. It also accommodates the different design of sleeve 300', which has only two fingers (332, 334) instead of a plurality of fingers. This design reduces complexity while maintaining functionality.

[0094] FIG. 16A is a cross-sectional view taken generally along line AA-AA in FIG. 13 and FIG. 16B is a cross-sectional view taken generally along line BB-BB in FIG. 13. Both figures illustrate dispensing apparatus 10'. Apparatus 10' comprises body 100', top cover 200', sleeve 300', actuator 600', bottom cover 400', and quick release attachment 500'. Apparatus 10' is designed to accommodate canister 12' therein.

[0095] In FIG. 16A, body 100' includes a third attachment aperture 114' proximate a top of body 100', a first attachment aperture 110' and a second attachment aperture 112' proximate a bottom of body 100', as generally shown in FIG. 14A. Top cover 200' includes fixed section 208' within third attachment aperture 114' and attachment aperture 314' of sleeve 300'. Top cover 200' also includes movable section 210' hinged connected to fixed section 208'. Sleeve 300' is nested within body 100'. Lower sleeve 800 is nested within body 100' and below sleeve 300'. Actuator 600' includes conduit 610' and dispensing opening 608'. Canister 12 includes nozzle 14. Bottom cover 400' is removably attached to body 100' proximate bottom 104' of body 100'. Quick release attachment 500' includes first extension 504', second extension 506', and body 502'.

[0096] FIG. 16B illustrates a similar view as in FIG. 16A, however, the cross-section cuts through first attachment aperture 110' proximate bottom 104' and second protrusion 118' of body 100'. First protrusion 116' is also shown. The other elements shown are the same as described in FIG. 16A, however, fewer features are shown due to the location of the cross-section.

[0097] It should be noted that bottom cover 400' may also include the components of bottom cover 400, as shown in FIG. 5. It should also be noted that quick-release attachment 500' is substantially similar to quick-release attachment 500 shown in FIG. 6 but it is designed to omit ring 516. It should be further noted that apparatus 10' could removably engage attachment 700, generally shown in FIGS. 10A through 11B, or alternatively, quick-release attachment 500' could be configured with some or all of the additional components of attachment 700.

[0098] FIG. 17 provides a partial front perspective view of the dispensing apparatus 10', with a portion cutaway to reveal the internal components and their interactions. The apparatus comprises body 100', top cover 200', sleeve 300', and actuator 600'. A portion of bottom cover 400' and quick

release attachment 500' (See FIGS. 12A and 12B) would be visible below body 100' in this view, but are omitted for clarity and because they do not directly interact with the components shown.

[0099] Body 100' is partially sectioned to show internal guide rails 128 and 130 and external guide rails 126 and 132. Internal guide rails 128 and 130 extend from curved portion 124 of the body and from an internal surface thereof and are configured to fit within channel 342 of sleeve 300'. External guide rails 126 and 132 are positioned out of the internal guide rails and extend higher up on the inner wall of body 100' and extends inwards, designed to interact with and bias first finger 332 and second finger 334 of sleeve 300' inwardly. The positioning of these guide rails helps to secure and align sleeve 300' within body 100'. Internal guide rails 130 and 128 are also visible within body 100'. These guide rails are shorter than guide rails 126 and 132, and work in conjunction with the other guide rails to stabilize, center, and restrict the vertical movement within the body of the internal components, e.g., actuator 600' and sleeve 300'.

[0100] Sleeve 300' is shown nested within body 100'. Channel 342, located near the bottom opening (304', not shown) of the sleeve, receives internal guide rails 130 and 128, further securing the sleeve's vertical position within the body and also preventing rotational movement with the body. First finger 332 and second finger 334 extend inwardly from the sidewall of sleeve 300' via external guide rails 132 and 126 and are each hingedly connected to the sleeve wall by living hinges, allowing them flexibility. These fingers are positioned close to the dispensing opening 608' of actuator 600'. The interaction of fingers 332 and 334 with external guide rails 132 and 126, creates a stop which restricts vertical movement of actuator 600', controlling the dispensing action. Leaf spring 320', attached to the upper portion of sleeve 300', extends upwards and is configured to interact with top cover 200', at least partially nested within depression 212'.

[0101] Top cover 200' includes fixed section 208' and movable section 210'. Movable section 210' includes depression 212', which receives at least a portion of leaf spring 320'. The leaf spring biases movable section 210' downwards, towards body 100', keeping top cover 200' in a closed position and until the user intentionally opens it. Top cover 200' is hingedly attached to body 100' at the top.

[0102] Actuator 600' is nested within sleeve 300' and its dispensing opening 608' is aligned with the dispensing aperture (108', not shown) of body 100' and dispensing opening 308' of sleeve 300'. When top cover 200' is open (shown in FIG. 18A and 18B and described further, infra), downward pressure on actuator 600' causes an aerosol container within sleeve 300' to dispense its contents. The cutaway view reveals the internal workings of the apparatus and how the various components interact to control the dispensing process.

[0103] FIGS. 18A and 18B offer partial cutaway views of dispensing apparatus 10', providing further detail on the arrangement and interaction of internal components, building upon the view shown in FIG. 17. Apparatus 10' comprises body 100', top cover 200', sleeve 300', lower sleeve 800, and actuator 600'. These figures highlight the interplay between the body's guide rails and the sleeve components, particularly the fingers of sleeve 300', where the interplay restricts vertical translational movement of actuator 600' within sleeve 300' and body 100'.

[0104] In FIG. 18A, a portion of body 100', top cover 200' and sleeve 300' is cut away to reveal finger 332 and sleeve 300'. Sleeve 300' is nested within body 100'. As shown in FIG. 17, first finger 332 and second finger 334, integral parts of sleeve 300', are biased inward by the interaction with external guide rails 132 and 126. This inward bias ensures that the fingers maintain contact with actuator 600', specifically shelf 626, further restricting its vertical movement when actuator 600' is pushed in a direction away from top cover 200'. Lower sleeve 800 is nested within body 100', adding a layer of stability and support for actuator 600' and also providing a housing for canister 10 therein. Leaf spring 320', attached to sleeve 300' near its top, extends upward and engages with depression 212' in top cover 200'. FIGS. 18A and 18B illustrate a substantially "open" position of top cover 200', specifically when upward force is imparted on bottom 204' of top cover 200'. Top cover 200', comprising fixed section 208' and movable section 210', is biased in closed position by leaf spring 320'. The angled edges (218', 220') of movable section 210' may contact edge 330' of sleeve 300' when the cover is closed.

[0105] FIG. 18B presents a similar view as FIG. 18A, but with a different section of body 100' removed to reveal external guide rail 132. This alternate cutaway provides a comprehensive view of the external guide rails, e.g., 132, bias the fingers, e.g., 332, radially inward within body 100'. The inclusion of shelf 626 on actuator 600' further restricts the actuator's downward movement within lower sleeve 800, sleeve 300', and body 100', as shelf 626 is designed to contact or abut one or more of the fingers of sleeve 300'.

[0106] It should be noted that the disclosed embodiments may be arranged in various combinations according to any of the embodiments shown and described. As such, the illustrated and described embodiments are intended to be illustrative and various alternatives, combinations, omissions, of specific components, or foreseeable alternative components, understood by one having ordinary skill in the art, described in the present disclosure or within the field of the present disclosure, are intended to fall within the scope of the appending claims.

[0107] It will be appreciated that various aspects of the invention and other features and functions, or alternatives thereof, may be desirably combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

LIST OF REFERENCE NUMERALS

[0108]	10 Apparatus
[0109]	10' Apparatus
[0110]	12 Canister
[0111]	14 Nozzle
[0112]	100 Body
[0113]	100' Body
[0114]	102 Top
[0115]	102' Top
[0116]	104 Bottom
[0117]	104' Bottom
[0118]	106 Internal space
[0119]	106' Internal space
[0120]	108 Discharge aperture
[0121]	108' Discharge aperture

[0122]	110 First attachment aperture	[0186]	330 Edge
[0123]	110' First attachment aperture	[0187]	330' Edge
[0124]	112 Second attachment aperture	[0188]	332 First finger
[0125]	112' Second attachment aperture	[0189]	334 Second finger
[0126]	114 Third attachment aperture	[0190]	336 Medial guide protrusion
[0127]	114' Third attachment aperture	[0191]	338 First side guide protrusion
[0128]	116 First protrusion	[0192]	340 Second side guide protrusion
[0129]	116' First protrusion	[0193]	342 Channel
[0130]	118 Second protrusion	[0194]	400 Bottom cover
[0131]	118' Second protrusion	[0195]	400' Bottom cover
[0132]	120 First groove	[0196]	402 Top
[0133]	122 Second groove	[0197]	404 Bottom
[0134]	124 Curved portion	[0198]	406 Channel
[0135]	126 First outer guide rail	[0199]	408 First protrusion
[0136]	128 First internal guide rail	[0200]	410 Second protrusion
[0137]	130 Second internal guide rail	[0201]	412 Aperture
[0138]	132 Second external guide rail	[0202]	414 First rim
[0139]	200 Top cover	[0203]	416 Second rim
[0140]	200' Top cover	[0204]	418 First channel
[0141]	202 Top	[0205]	420 Second channel
[0142]	204 Bottom	[0206]	500 Quick-release attachment
[0143]	204' Bottom	[0207]	500' Quick-release attachment
[0144]	206 Hinge	[0208]	502 Body
[0145]	206' Hinge	[0209]	502' Body
[0146]	208 Fixed section	[0210]	504 First extension
[0147]	208' Fixed section	[0211]	504' First extension
[0148]	210 Movable section	[0212]	506 Second extension
[0149]	210' Movable section	[0213]	506' Second extension
[0150]	212 Depression	[0214]	508 First protruding tab
[0151]	212' Depression	[0215]	508' First protruding tab
[0152]	214 First stop	[0216]	510 Second protruding tab
[0153]	216 Second stop	[0217]	510' Second protruding tab
[0154]	218 First angled edge	[0218]	512 Housing
[0155]	218' First angled edge	[0219]	512' Housing
[0156]	220 Second angled edge	[0220]	514 Aperture
[0157]	220' Second angled edge	[0221]	514' Aperture
[0158]	300 Sleeve	[0222]	516 Keyring
[0159]	300' Sleeve	[0223]	518 Slot
[0160]	301 Sidewall	[0224]	600 Actuator
[0161]	301' Sidewall	[0225]	600' Actuator
[0162]	302 Top	[0226]	602 Top
[0163]	302' Top	[0227]	602' Top
[0164]	304 Bottom	[0228]	604 Bottom
[0165]	304' Bottom	[0229]	604' Bottom
[0166]	306 Internal space	[0230]	606 Conduit
[0167]	306' Internal space	[0231]	606' Conduit
[0168]	308 Second discharge aperture	[0232]	608 Dispensing opening
[0169]	308' Second discharge aperture	[0233]	608' Dispensing opening
[0170]	310 First plurality of fingers	[0234]	610 Inflow opening
[0171]	312 Second plurality of fingers	[0235]	610' Inflow opening
[0172]	314 Fourth attachment aperture	[0236]	612 First plurality of slots
[0173]	314' Attachment aperture	[0237]	614 Second plurality of slots
[0174]	316 Top surface	[0238]	616 Top surface
[0175]	316' Top surface	[0239]	616' Top surface
[0176]	318 Hinge	[0240]	618 Sidewall
[0177]	318' Hinge	[0241]	618' Sidewall
[0178]	320 Leaf spring	[0242]	620 Medial guide depression
[0179]	302' Leaf spring	[0243]	622 First side guide depression
[0180]	322 First extension	[0244]	624 Second side guide depression
[0181]	324 Second extension	[0245]	626 Shelf
[0182]	326 Protrusion	[0246]	700 Quick-release attachment
[0183]	326' Protrusion	[0247]	702 Body
[0184]	328 Protrusion	[0248]	704 First extension
[0185]	328' Protrusion	[0249]	706 Second extension

[0250] 708 First protruding tab
 [0251] 710 Second protruding tab
 [0252] 712 Housing
 [0253] 714 Aperture
 [0254] 718 Slot
 [0255] 720 Clip extension
 [0256] 722 Clip body
 [0257] 724 Arcuate portion
 [0258] 726 End
 [0259] 728 Surface
 [0260] 800 Lower sleeve

What is claimed is:

1. A dispensing apparatus comprising:
 - a body having an open top section and a bottom section, further comprising:
 - at least one dispensing aperture and a first attachment aperture proximate the top section;
 - a second and third attachment aperture proximate the bottom section;
 - a leaf spring hingedly attached to the top proximate the first attachment aperture;
 - an actuator having a conduit defined by a dispensing opening and an inflow opening slidably arranged within the body;
 - a top cover having a fixed section frictionally secured within the first attachment aperture and a movable section hingedly connected to the fixed section;
 - a bottom cover rotatably secured to the bottom section of the body; and,
 - a quick-release attachment arranged to removably secure to the second and third attachment apertures;

wherein an aerosol container is arranged to be removably secured within the body and dispenses contents from the dispensing aperture from the dispensing opening when the actuator is pushed towards the bottom section within the sleeve.

2. The dispensing apparatus recited in claim 1 further comprising:

- a sleeve arranged within the body, the sleeve having an open top section and a bottom section, the leaf spring hingedly attached proximate the top section, the sleeve having a second dispensing aperture, the sleeve having a fourth attachment aperture proximate the top section, wherein the second dispensing aperture is arranged to be in communication with the at least one dispensing aperture and the fourth attachment aperture is arranged to be substantially colinear with the first attachment aperture of the body when the sleeve is arranged within the body, wherein the actuator and the aerosol container are arranged within the sleeve.

3. The dispensing apparatus recited in claim 2, wherein the sleeve further comprises a first plurality of fingers and a second plurality of fingers, each of the fingers hingedly connected to the sleeve, wherein both of the plurality of fingers restrict downward slidable movement of the actuator within the sleeve.

4. The dispensing apparatus recited in claim 2, wherein the sleeve further comprises a first finger and a second being disposed on opposite sides of the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.

5. The dispensing apparatus recited in claim 4, wherein the body includes a first external guide rail and a second external guide rail disposed on an internal surface thereof,

the external guide rails adapted to bias each of the fingers of the sleeve in a radially inward direction when the sleeve is within the body.

6. The dispensing apparatus recited in claim 5, wherein the body further includes a first internal guide rail and a second internal guide rail disposed on an internal surface thereof and arranged between the external guide rails, the internal guide rails arranged to rest at least partially within a channel of the sleeve, thereby restricting downward movement of the sleeve within the body.

7. The dispensing apparatus recited in claim 1, wherein the leaf spring biases the movable section of the top cover downwardly thereby maintaining the movable section above the actuator.

8. The dispensing apparatus recited in claim 1, wherein the quick-release attachment comprises a key ring extending therefrom, the quick-release attachment having a pair of protrusions arranged to frictionally and removably engage the second and third attachment apertures.

9. The dispensing apparatus recited in claim 1, wherein the quick-release attachment comprises a clip extending therefrom and arranged to rest on an outer surface of the body, the quick-release attachment having a pair of protrusions arranged to frictionally and removably engage the second and third attachment apertures.

10. A dispensing apparatus, comprising:

- a hollow body having a top opening and a bottom opening, the body including a discharge aperture proximate the top opening;

- an actuator slidably arranged within the body, the actuator having a conduit therein, the conduit having a first opening in communication with the discharge aperture and a second opening proximate the bottom;

- a top cover secured to the body proximate the top, the cover having a movable section hingedly attached to the body;

- a leaf spring extending from the top and arranged to substantially abut a top surface of the movable section, thereby applying a downward force thereon; and,

- a bottom cover removably secured to the bottom;

wherein the actuator is arranged to actuate an aerosol container within the body when downward force is applied to the actuator, thereby applying downward force to a nozzle of the aerosol container and discharging contents of the aerosol container through the conduit and out of the discharge aperture.

11. The dispensing apparatus recited in claim 10, wherein the leaf spring biases the movable section in a downward direction when the movable section is moved by a user in an upward direction, thereby substantially covering the actuator unless constant upward force is applied to the movable section.

12. The dispensing apparatus recited in claim 10 further comprising:

- a quick-release attachment, the attachment having a pair of attachment tabs extending therefrom, the attachment tabs removably and frictionally engaged to a respective pair of attachment apertures arranged proximate the bottom opening of the body.

13. The dispensing apparatus recited in claim 12, wherein the quick release-attachment includes at least one of:

- a key-ring; and,
- a spring clip.

14. The dispensing apparatus recited in claim **11** further comprising:

a hollow sleeve having a top opening and bottom opening, the sleeve having a second discharge aperture therein, the sleeve including one of:

at least a first plurality of fingers hingedly secured thereon, wherein the sleeve is arranged to be inserted within the body and the actuator is arranged to be inserted within the sleeve, wherein at least one finger of the first plurality of fingers restricts translational movement of the actuator in a direction of the bottom opening of the sleeve; or,

a first finger and a second being disposed on opposite sides of and proximate the second dispensing aperture, each of the fingers hingedly connected to the sleeve, wherein both of the fingers restrict downward slidable movement of the actuator within the sleeve.

15. The dispensing apparatus recited in claim **14**, wherein the body includes a cover aperture proximate the top opening thereof and the sleeve includes a cover aperture proximate the top opening thereof, wherein the top cover includes a fixed portion arranged to be frictionally secured at least partially within the cover apertures, thereby securing the sleeve within the body.

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